LATENT STATE SPECTRAL TRANSFER: SUPPLEMENTARY MATERIAL

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1 SUPPLEMENTARY RESULTS AND DERIVATIONS

This supplement gives the derivations, analytical sweeps, learned-representation measurements, intervention boundaries, and temporal evaluation protocols used in the main manuscript. Sections 2–4 derive and test the same-spectrum commutator identity; Sections 5–8 separate alignment from spectral shape and examine robustness; and Section 10 provides an additional spectrum-fixed subspace diagnostic. Sections 11–12 analyze frozen LeWorldModel representations and the prediction consequences of alignment interventions, including learned pixel-to-latent, equivariance, and OOD/planning boundary benchmarks. Section 13 analyzes frozen I-JEPA representations, while Section 14 evaluates released JEPA-WM and DINO-WM checkpoints with held-out fits, action-order interventions, split perturbations, and estimator ablations.

The formal result concerns the finite-dimensional same-spectrum construction: eigenvalues do not determine covariance–predictor eigenspace alignment. The learned-representation experiments evaluate the corresponding statistics outside that construction without extending the theorem's assumptions. Section 15.9 then separates simulator action effects, frozen-model counterfactual sensitivity, and live closed-loop planning rather than treating any one of them as a substitute for the others.

2 COMPLETE MATHEMATICAL DERIVATION

Let $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_d)$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$. Let R_θ rotate only coordinates (i, j) and define $P_\theta = R_\theta \Lambda R_\theta^\top$. Since orthogonal conjugation preserves eigenvalues, every P_θ has the same predictor spectrum. The only entries of P_θ that change in a way relevant to $[\Sigma, P_\theta]$ lie in the rotated two-dimensional block.

Write $c = \cos\theta$, $s = \sin\theta$, $\Delta_\sigma = \sigma_i - \sigma_j$, and $\Delta_\lambda = \lambda_i - \lambda_j$. The rotated block is

$$P_{\theta,ij} = \begin{bmatrix} \lambda_i c^2 + \lambda_j s^2 & \Delta_\lambda cs \\ \Delta_\lambda cs & \lambda_i s^2 + \lambda_j c^2 \end{bmatrix}. \tag{S1}$$

Multiplying Eq. (S1) by the diagonal block of Σ on the left and right gives

$$\Sigma_{ij} P_{\theta,ij} - P_{\theta,ij} \Sigma_{ij} = \begin{bmatrix} 0 & \Delta_\sigma \Delta_\lambda cs \\ -\Delta_\sigma \Delta_\lambda cs & 0 \end{bmatrix}. \tag{S2}$$

Thus

$$\|[\Sigma, P_\theta]\|_F^2 = 2(\Delta_\sigma \Delta_\lambda cs)^2 = \frac{(\Delta_\sigma \Delta_\lambda)^2 \sin^2(2\theta)}{2}. \tag{S3}$$

By Eq. (S3), the normalized commutator has the same angular dependence because $\|\Sigma\|_F$ is fixed and $\|P_\theta\|_F = \|\Lambda\|_F$ for every θ. The falsifiers follow directly: if $\Delta_\sigma = 0$ or $\Delta_\lambda = 0$ on the rotated plane, the commutator is zero at all angles.

The same block calculation gives the horizon-one coordinate energy used in Proposition 2. If Π_i projects onto the first coordinate of the rotated plane, then the first row of $P_{\theta,ij}$ is $(\lambda_i c^2 + \lambda_j s^2, \Delta_\lambda cs)$. Therefore

$$\begin{aligned} \mathrm{tr}(\Pi_i P_\theta \Sigma P_\theta^\top) = \sigma_i (\lambda_i c^2 + \lambda_j s^2)^2 \\ + \sigma_j (\Delta_\lambda cs)^2. \end{aligned} \tag{S4}$$

Equation (S4) is not determined by the unordered predictor eigenvalue list; it depends on how the predictor eigenvectors sit relative to the covariance coordinates.

Table 1: Exact commutator curve used by the main theorem. The construction rotates coordinates 0 and 63, with latent eigengap 0.984375 and predictor eigengap 0.875.

Rotation angle	Exact normalized commutator $\mathcal{C}$
0°	0.000000
15°	0.109531
30°	0.189714
45°	0.219063
60°	0.189714
75°	0.109531
90°	0.000000

3 Main Same-Spectrum Rotation Sweep

The sweep holds the predictor eigenvalues fixed and changes only the predictor basis. It is the direct test of the spectrum-only explanation: the spectrum is invariant, whereas the commutator and transfer readouts are free to change.

Table 2: Complete same-spectrum rotation sweep. The predictor eigenvalues are fixed across rows; the table reports covariance–predictor mismatch and transfer readouts for each basis rotation.

Rotation angle	Normalized commutator $\mathcal{C}$	$h = 1$ top-band lift	$h = 1$ residual-band lift	Residual ratio $\mathcal{E}_8/\mathcal{E}_1$
0°	0.000000	1.98127	1.39623	2.23004
15°	0.109531	1.91896	1.35963	2.20256
30°	0.189714	1.76871	1.30747	2.13518
45°	0.219063	1.63077	1.32836	2.05798
60°	0.189714	1.62753	1.43086	1.99393
75°	0.109531	1.78847	1.54632	1.95356
90°	0.000000	1.90720	1.59578	1.93994

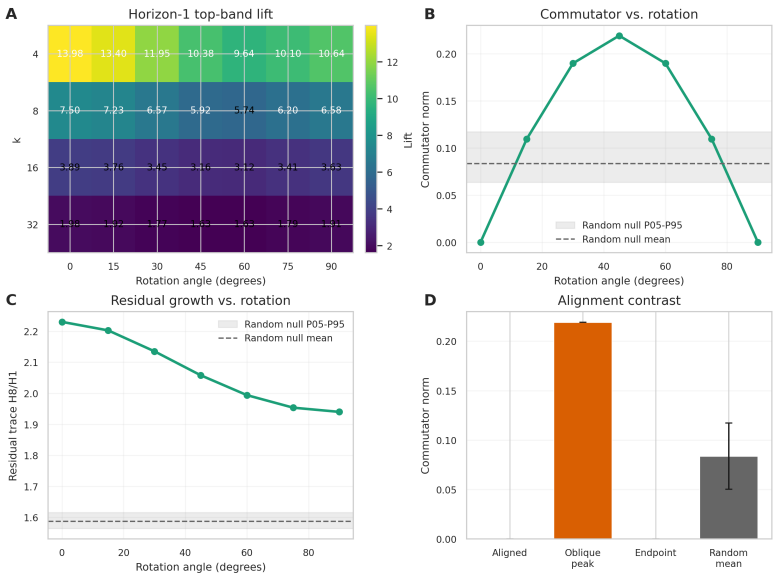


Figure 1: Main commutator and eigenspace-alignment figure. The same-spectrum rotation changes commutator, top-band lift, and residual growth.

4 NULL COMPARISONS

The null suite distinguishes the proposed commutator identity from two nearby explanations. The permutation-preserving null keeps the pair jointly diagonal, so it tests whether coordinate reordering alone can create the signal. The random-basis null keeps the predictor spectrum but removes the structured single-plane perturbation, so it tests whether any spectrum-matched basis is sufficient.

Table 3: Spectrum-preserving null sweep. The oblique construction is compared with a jointly diagonal permutation control and a spectrum-matched random-basis control. Larger $\mathcal{C}$ indicates greater covariance–predictor eigenspace mismatch.

Basis construction	Mean $\mathcal{C}$	5th percentile	95th percentile
Oblique plane rotation	0.219063	0.219063	0.219063
Permutation-preserving null	0.000000	0.000000	0.000000
Random-basis null	0.084512	0.061633	0.121488

The random-basis distribution from the main sweep has commutator mean 0.0836631 and a 5th–95th percentile interval of $[0.0637529, 0.11723]$; its residual-growth and predicted-decay means are 1.58794 and 0.211866. These controls are intentionally nonzero. Random bases can create noncommutativity, but they do not reproduce the structured peak generated by the single-plane rotation.

5 FACTOR-ISOLATING ABLATIONS

The main ablation separates two variables that a spectrum-only summary would conflate. The first block fixes spectrum and varies alignment. The second fixes alignment at $45°$ and varies input spectral anisotropy. This establishes that spectrum shape affects effect size, while basis alignment remains an independent descriptor of covariance-predictor action.

Table 4: Decoupling predictor spectrum from covariance–predictor eigenspace alignment. The two blocks vary one factor while holding the other fixed.

Isolation axis	Perturbation	Latent stable rank	Normalized commutator $\mathcal{C}$	$h = 1$ top-band lift	Residual ratio $\mathcal{E}_8/\mathcal{E}_1$
Fixed spectrum	$0°$ angle	4.74389	0.000000	1.98127	2.23004
Fixed spectrum	$45°$ angle	4.74389	0.219063	1.63077	2.05798
Fixed spectrum	$90°$ angle	4.74389	0.000000	1.90720	1.93994
Fixed alignment	exponent 0.0	14.60210	0.114121	1.65769	1.78340
Fixed alignment	exponent 0.5	7.89448	0.176617	1.65943	1.91872
Fixed alignment	exponent 1.0	4.74389	0.219063	1.63077	2.05798
Fixed alignment	exponent 1.5	3.18365	0.244025	1.58815	2.16180
Fixed alignment	exponent 2.0	2.36335	0.258604	1.54226	2.19337

Table 5: Robustness of transfer readouts to the chosen energy band and horizon. The same rotation grid is evaluated with four readouts.

Rotation angle	$h = 1$ top-band lift	$h = 1$ residual-band lift	$h = 8$ top-band lift	$h = 8$ residual-band lift
$0°$	1.98127	1.39623	2.00000	1.62670
$15°$	1.91896	1.35963	1.86631	1.60092
$30°$	1.76871	1.30747	1.50133	1.55792
$45°$	1.63077	1.32836	1.00394	1.55638
$60°$	1.62753	1.43086	0.511423	1.61080
$75°$	1.78847	1.54632	0.178818	1.68093
$90°$	1.90720	1.59578	0.227794	1.71232

6 Anisotropy and Dimension Boundaries

The anisotropy scan changes the input spectral exponent continuously while holding the oblique angle fixed. The peak angle remains 45° in the tested grid, while peak commutator grows as stable rank decreases.

Table 6: Continuous sweep over input spectral anisotropy.

Input exponent	Latent stable rank	Peak normalized commutator $\mathcal{C}$	Peak top-band lift
0.0	14.60210	0.114121	1.00506
0.5	7.89448	0.176617	1.00457
1.0	4.74389	0.219063	1.00394
1.5	3.18365	0.244025	1.00334
2.0	2.36335	0.258604	1.00281

Table 7: Cross-dimension theory sweep. The ordering induced by predictor spectral decay is evaluated at three finite dimensions.

Dimension	Baseline $\mathcal{E}_8/\mathcal{E}_1$	$\beta = 0.5$ $\mathcal{E}_8/\mathcal{E}_1$	$\beta = 2.0$ $\mathcal{E}_8/\mathcal{E}_1$	Ordering preserved?	Tail gap	Heat-recovery R^2
64	2.23004	4.56383	1.39186	Yes	1.0	0.999922
128	1.98384	3.85419	1.31785	Yes	1.0	0.999977
256	1.81068	3.34861	1.26605	Yes	1.0	0.999953

7 Mechanism Sweep

The mechanism sweep tests how residual growth changes when the analytical generator varies encoder and predictor spectral response. It reports residual growth, predicted decay, and tail-band residual concentration under the same deterministic construction used by the main experiments.

Table 8: Interaction between encoder and predictor spectral response. Each row reports the resulting residual growth and predicted-trace ratio.

Encoder exponent	Predictor exponent	Latent spectral exponent	Residual ratio $\mathcal{E}_8/\mathcal{E}_1$	Predicted-trace ratio $\mathcal{P}_8/\mathcal{P}_1$
0.5	0.5	0.75	3.33115	0.231924
0.5	1.0	0.75	1.73926	0.306679
0.5	2.0	0.75	1.21394	0.520081
1.0	0.5	1.00	3.85419	0.352804
1.0	1.0	1.00	1.98384	0.425678
1.0	2.0	1.00	1.31785	0.612044
1.5	0.5	1.25	4.58154	0.475850
1.5	1.0	1.25	2.32305	0.534648
1.5	2.0	1.25	1.45725	0.686855
2.0	0.5	1.50	5.53159	0.584591
2.0	1.0	1.50	2.75917	0.626212
2.0	2.0	1.50	1.62741	0.746749

Table 9: Sensitivity to the synthetic heat filter. The table reports retained trace, tail-band mass, and recovery of the known generator.

Heat exponent	Diffusion time	Retained trace fraction	Retained tail fraction	Generator-recovery R^2
0.5	0.000	1.000000	0.126861	0.960860
0.5	0.001	0.992239	0.125425	0.960860
0.5	0.003	0.976993	0.122591	0.960860
0.5	0.010	0.926403	0.113063	0.960860
0.5	0.030	0.802470	0.089073	0.960860
1.0	0.000	1.000000	0.126861	0.999792
1.0	0.001	0.955755	0.110312	0.999792
1.0	0.003	0.881913	0.082903	0.999792
1.0	0.010	0.717684	0.029450	0.999792
1.0	0.030	0.523308	0.001538	0.999792
2.0	0.000	1.000000	0.126861	0.127313
2.0	0.001	0.625067	0.000004	0.127313
2.0	0.003	0.524025	0.000000	0.127313
2.0	0.010	0.413441	0.000000	0.127313
2.0	0.030	0.312955	0.000000	0.127313

Table 10: Oblique-frame stress test. Increasing obliqueness changes finite-window slope bias and the frame condition number.

Obliqueness	Baseline mean absolute bias	Broken-tail mean absolute bias	Frame condition number
0.00	0.000000	0.000000	1.00000
0.25	0.129676	0.072982	1.63916
0.50	0.415790	0.262480	2.71704
0.75	0.759680	0.517336	4.54002
1.00	1.12440	0.810285	7.64852

8 Robustness Sweeps

The robustness sweep varies random seeds for auxiliary frame diagnostics and adds small perturbations to the heat-recovery experiment. These measurements check whether the reported diagnostic quantities are stable under nuisance variation in the analytical setup.

Table 11: Seed robustness of oblique-bias and frame diagnostics. Entries summarize nuisance draws in the analytical setup.

Diagnostic quantity	Mean	Standard deviation	Minimum	Maximum
Baseline oblique bias	0.432336	0.009375	0.415378	0.441991
Broken-tail oblique bias	0.284172	0.011879	0.262480	0.303541
Frame condition number	2.61823	0.048972	2.56380	2.71704

Table 12: Recovery of the known heat generator under additive noise.

Noise standard deviation	Mean R^2	Minimum R^2	Maximum R^2
0.00	1.000000	1.000000	1.000000
0.01	0.999980	0.999977	0.999983
0.03	0.999819	0.999790	0.999849
0.05	0.999498	0.999416	0.999580

9 DIAGONAL-TRANSFER SPECTRAL DIAGNOSTICS

This section reports diagonal-transfer diagnostics that complement the same-spectrum construction. The sweep varies predictor decay and broken-tail spectral structure, then reports residual growth, fitted residual decay, and heat-recovery accuracy under the analytical generator.

Table 13: Diagonal-transfer spectral diagnostics. Rows vary predictor decay or introduce broken-tail spectral structure under the analytical generator.

Case	Latent exponent α	Tail gap	Residual exponent	Residual ratio $\mathcal{E}_8/\mathcal{E}_1$	Heat-recovery R^2
Baseline	1.00000	0.000000	0.691532	1.98384	0.999977
Predictor 0.5	1.00000	0.000000	0.538399	3.85419	0.999977
Predictor 2.0	1.00000	0.000000	0.861305	1.31785	0.999977
Broken tail 3 b32	1.44847	1.00000	1.18434	2.19174	0.999977
Broken tail 4 b64	1.25910	1.50000	0.979362	2.06464	0.999977

10 AUXILIARY SUBSPACE-ROTATION DIAGNOSTIC

This auxiliary sweep checks a second spectrum-fixed rotation diagnostic. It measures how rotating the observed subspace changes latent mass, residual energy, and top-band lift under the same synthetic construction.

Table 14: Auxiliary subspace-rotation sweep. The spectrum is fixed while the measured subspace is rotated.

Rotation angle	Latent-mass share	Residual-energy share	Top-32 lift
0°	0.572917	0.249719	2.00000
15°	0.536409	0.283857	1.86603
30°	0.436667	0.377125	1.50000
45°	0.300416	0.504532	1.00000
60°	0.164165	0.631938	0.50000
75°	0.064422	0.725206	0.133975
90°	0.027914	0.759345	0.000000

The random null for this diagnostic has top-32 lift mean 1.01115 with a 5th–95th percentile interval from 0.927697 to 1.09239. Its residual-share mean is 0.616699 and latent-mass-share mean is 0.127282. The structured rotation therefore changes both top-band mass and residual share relative to the random-subspace reference.

11 FROZEN PUSHT LATENT MEASUREMENTS

The theory construction is synthetic by design, but the same measurement vocabulary can be applied to frozen learned latents. This section reports an offline measurement on frozen PushT latents with top-band size $k = 32$. The readouts compare observed latent directions with randomized controls that preserve scale or randomize basis structure.

Table 15: Frozen PushT top-band readouts. Observed top-32 lift is compared with a norm-matched randomized control; ratios above one indicate greater concentration in the top covariance band than in the reference.

Readout	Observed top-band lift	Randomized-reference lift	Observed/reference ratio
Dynamics	4.71084	1.00082	4.70698
Planner-weighted	4.62362	1.00082	4.61983
Predicted dynamics	4.68678	1.00082	4.68294
Prediction residual	3.68668	1.00082	3.68366

Table 16: Frozen PushT label-shuffle controls. Shuffles remove action labels, state-cluster labels, or temporal ordering while retaining the same frozen latent records and top-32 readout.

Readout	Observed ordering	Action-label shuffle	State-label shuffle	Temporal-order shuffle
Dynamics	4.71084	1.86731	1.86817	2.18068
Planner-weighted	4.62362	1.94846	1.94945	2.14740
Predicted dynamics	4.68678	1.83563	1.83670	2.16529
Prediction residual	3.68668	1.96542	1.96579	2.05093

Table 17: Frozen PushT transition-overlap distribution. Observed transitions are compared with a random-subspace reference using the same top-band size $k = 32$.

Statistic	Observed transitions	Random-subspace reference
Mean overlap	0.029970	0.020953
5th-percentile overlap	0.004339	0.010673
95th-percentile overlap	0.111357	0.034016

Table 18: Frozen PushT conditioning sweep. The same top-32 readouts are recomputed under state-, action-, latent-, and global-conditioned summaries.

Conditioning summary	Dynamics	Predicted dynamics	Prediction residual	Planner-weighted dynamics
States	4.67541	4.64709	3.66198	4.59063
Actions	2.83562	2.82753	2.42784	2.97068
Latents	5.29517	5.27070	4.00016	5.18173
Global	1.83645	1.79945	1.95409	1.90871

To test whether the state-conditioned action readout is an in-sample artifact, we independently split each of the eight state cells into two halves, fit a zero-intercept linear map from actions to dynamics on one half, and evaluate it on the other half. For a test index set I_{test}, the reported score is

$$R^2_{\text{test}} = 1 - \frac{\sum_{i \in I_{\text{test}}} \|\Delta z_i - a_i \widehat{B}_{\text{train}}\|_2^2}{\sum_{i \in I_{\text{test}}} \|\Delta z_i - \overline{\Delta z}_{\text{test}}\|_2^2}, \tag{S5}$$

where $\widehat{B}_{\text{train}}$ is estimated only on the complementary training half. The permutation null shuffles action rows within the sampled artifact before applying the same split and estimator.

Table 19: State-conditioned action-to-dynamics readout with an independent held-out and permutation-null check. Each seed uses 20,000 deterministic rows, eight state cells, and a within-cell half/half split. The primary fit has no intercept; the score is descriptive and is not a controllability or planning metric. The final column counts cells whose $k = 32$ dynamics-minus-controllable lift is non-positive.

Initialization replicate	Pooled in-sample R^2	Held-out cell R^2 mean	Held-out cell range	Permuted-action null	Non-positive cells / 8
3072	0.03198	0.06034	0.04524–0.07535	−0.02422	8
3073	0.03121	0.05593	0.04449–0.08024	−0.03469	8
3074	0.03194	0.05502	0.04194–0.07600	−0.03460	8

The held-out cell-level scores are positive for all three seeds and exceed the corresponding action-permutation nulls. The in-sample state-cell means are also above the pooled in-sample readout, but the held-out check is the more relevant guard against a same-sample fit artifact. No cell is flagged by the report's operational high-variance/low-controllability diagnostic, and all 24 cells have non-positive $k = 32$ dynamics-minus-controllable lift. These checks support the narrow statement that action information is detectable in the frozen latent dynamics under state conditioning. They do not establish a causal controllability mechanism, goal reachability, physical observability, online planning, or multi-step rollout quality; the export lacks goals, success, episode/timestep identifiers, physical variables, counterfactual transitions, and multi-horizon dynamics.

These learned-latent measurements show that basis-sensitive statistics move relative to randomized controls. The mean transition overlap is $1.43\times$ the random-subspace mean, the observed upper tail is

wider than the null interval, and planner weights put $1.44\times$ more mass on the top latent band than on the tail band. Recomputing the state-conditioned readouts on the two stored sample sizes changes the dynamics lift from 4.67541 to 4.71084; the corresponding ratios for predicted dynamics, residual, and planner-weighted readouts are 0.99153, 0.99330, and 0.99287. The stable ordering across these records indicates that the readout is not determined by a single sample cap, while the absolute value remains a property of the sampling protocol.

The PushT measurements show that the alignment-sensitive readouts are nontrivial in frozen learned latents. Because the stored records support offline one-step analysis, these results characterize representation geometry rather than online rollouts, counterfactual transitions, or task success.

12 Alignment Interventions and Prediction Boundaries

The diagnostic is not itself a claim that lower covariance–predictor mismatch must improve prediction. We therefore add two matched intervention studies. The first is a learned synthetic benchmark in which two action groups have different covariance and transition spectra. Each local covariance–transition pair commutes (the exact local normalized commutators are 0 and 5.68×10^{-16}), whereas the pooled data-generating commutator is 0.354281. An identical action-conditioned linear predictor is trained with no alignment, global alignment, or action-conditional alignment. All rows use the same generated splits, initialization seeds, optimizer, and 300-epoch training protocol. The rollout metric repeats each sample's action label at every step, so it is a repeated-action stability diagnostic rather than an arbitrary action-sequence evaluation. Ordered two-action composition is measured separately for $0\to0$, $0\to1$, $1\to0$, and $1\to1$.

Table 20: Learned synthetic intervention matrix. The local data-generating operators commute, but the pooled geometry is heterogeneous. Values are means over three seeds; the one-step MSE and $h = 8$ rollout MSE are evaluated on a held-out split and the rollout repeats the same action label. Composition error is the mean normalized error over the four ordered two-action products. The experiment is a mechanism boundary test, not a learned LeWorldModel result.

Mode	Weight	Test one-step MSE	$h = 8$ rollout MSE	Composition error	Global $\mathcal{C}$	Conditional $\mathcal{C}$
None	0	2.45792×10^{-6}	1.93476×10^{-6}	0.00815304	0.355696	3.22857×10^{-5}
Global	0.01	2.19856×10^{-5}	1.78173×10^{-5}	0.0146905	0.351926	5.84717×10^{-5}
Global	0.1	3.98326×10^{-3}	4.30374×10^{-3}	0.361108	0.288497	2.25252×10^{-3}
Global	1	8.07862×10^{-2}	3.80981×10^{1}	3.78536	0.0740325	0.019993
Global	10	0.290907	0.509461	1.63435	0.0033793	0.295549
Global	100	0.350466	0.492963	1.38404	0.000430968	0.269849
Conditional	0.01	2.24614×10^{-6}	1.73940×10^{-6}	0.00756084	0.355684	2.65222×10^{-5}
Conditional	0.1	2.35859×10^{-6}	1.59203×10^{-6}	0.00790262	0.355676	3.15303×10^{-5}
Conditional	1	8.22803×10^{-6}	4.99238×10^{-6}	0.00913680	0.355836	4.59051×10^{-5}
Conditional	10	2.14002×10^{-3}	9.13679×10^{-5}	0.149069	0.351436	6.80257×10^{-5}
Conditional	100	0.0575382	0.0205186	1.51700	0.265642	1.72814×10^{-4}

The synthetic matrix establishes a necessary boundary for interpreting the regularizer. Conditional alignment preserves the local commutator at small weight and remains close to the no-alignment predictor, whereas global alignment imposes an inappropriate pooled geometry and degrades prediction. At larger conditional weights, the alignment diagnostic can continue to fall while prediction error rises. Thus commutator reduction is an intervention target, not a sufficient objective for predictive quality.

We next test the same distinction on real PushT records exported from a frozen LeWorldModel checkpoint. This is a post-hoc latent adapter rather than end-to-end retraining. Given the frozen displacement $\widehat{\Delta z}$, the adapter predicts

$$\widehat{z}_{t+1}^{\mathrm{adapt}} = z_t + \widehat{\Delta z}_{t+1} + g_\phi\Big(z_t, \widehat{\Delta z}_{t+1}, a_t\Big), \tag{S6}$$

where g_ϕ is initialized at zero and is trained against the exported one-step displacement. The experiment uses 100,000 deterministic rows, an 80/10/10 row-level split, ten epochs, and three initialization seeds. Because the compact export does not contain episode or timestep identifiers, only held-out one-step prediction is evaluated.

Table 21: Frozen PushT latent-adapter intervention. The frozen row is the released predictor without an adapter; all other rows are the same learned residual adapter with the indicated alignment penalty. Paired MSE deltas are computed within seed relative to the no-alignment adapter.

Intervention	Weight	Held-out MSE	Paired ΔMSE vs. none	Conditional $\mathcal{C}$
Frozen predictor	–	0.00776849	–	0.00921481
Adapter, none	0	0.00622429	0	0.00913191
Adapter, global	0.001	0.00622426	-2.28×10^{-8}	0.00913102
Adapter, global	0.01	0.00622410	-1.89×10^{-7}	0.00912299
Adapter, global	0.1	0.00622562	$+1.33 \times 10^{-6}$	0.00904336
Adapter, conditional	0.001	0.00622427	-1.37×10^{-8}	0.00913080
Adapter, conditional	0.01	0.00622419	-9.86×10^{-8}	0.00912071
Adapter, conditional	0.1	0.00622677	$+2.48 \times 10^{-6}$	0.00902090

The residual adapter itself supplies the main one-step improvement over the frozen predictor. Relative to that matched adapter baseline, alignment changes the conditional diagnostic but has only a negligible effect at weight 0.01; the stronger conditional penalty lowers $\mathcal{C}$ further while worsening MSE. These real-record results therefore support a narrower conclusion: alignment is an identifiable intervention on the latent geometry, but its predictive value is conditional on the weighting and cannot be inferred from the diagnostic alone. They do not establish end-to-end training gains, multi-step rollout gains, or online planning success.

12.1 External Multi-step Adapter and Normalization Robustness

The preceding row-level adapter does not test whether a geometry intervention survives autoregressive rollout. We therefore fit a second, deliberately small residual adapter on one-step transitions from the public external PushT archive and evaluate it on held-out episodes. The LeWorldModel encoder and predictor remain frozen. There are 21 episodes and 42 windows; three episode split seeds (3072–3074) and three adapter initialization seeds are used. The alignment weights are 0.01, 0.1, and 1.0, and the two modes are the global and action-conditional penalties. The public action normalization is repeated with validation-split z-scoring and raw actions divided by 100. All conditions use the same checkpoint-selection rule and held-out rollout code. The independent unit is the episode split seed, with initialization seeds nested inside it.

Table 22 reports the primary $h = 8$ contrast, defined as adapted minus matched no-alignment rollout MSE. Negative values favor the adapter. The public-normalization global condition is consistently positive at $h = 8$, whereas the conditional condition is mostly compatible with zero. Under the other normalizations, signs and magnitudes change with the penalty weight. Thus a measurable alignment intervention does not imply a monotone multi-step gain.

Table 22: External frozen temporal adapter robustness at $h = 8$. Values are split-level paired MSE deltas (adapter minus no-alignment) with 95% episode bootstrap intervals. Negative values favor the adapter.

Normalization	Mode	Weight	ΔMSE vs. none	95% interval
public	global	0.01	$+0.006871$	$[+0.000046, +0.020366]$
public	global	0.1	$+0.007218$	$[+0.003033, +0.014978]$
public	global	1	$+0.027554$	$[+0.001312, +0.075299]$
public	conditional	0.01	$+0.000450$	$[-0.000146, +0.001206]$
public	conditional	0.1	$+0.002983$	$[-0.002650, +0.006824]$
public	conditional	1	$+0.003585$	$[-0.011599, +0.026681]$
validation z-score	global	0.01	-0.000079	$[-0.000824, +0.000586]$
validation z-score	global	0.1	-0.000847	$[-0.006940, +0.002486]$
validation z-score	global	1	-0.005210	$[-0.022712, +0.007698]$
validation z-score	conditional	0.01	$+0.000169$	$[-0.001300, +0.001603]$
validation z-score	conditional	0.1	$+0.000895$	$[-0.008999, +0.009866]$
validation z-score	conditional	1	-0.037889	$[-0.123547, +0.038389]$
raw/100	global	0.01	$+0.000122$	$[+0.000046, +0.000202]$
raw/100	global	0.1	$+0.000155$	$[-0.000530, +0.000957]$
raw/100	global	1	$+0.000553$	$[-0.002457, +0.004996]$
raw/100	conditional	0.01	$+0.000219$	$[+0.000123, +0.000312]$
raw/100	conditional	0.1	$+0.000727$	$[+0.000207, +0.001428]$
raw/100	conditional	1	$+0.003174$	$[-0.001608, +0.008897]$

12.2 Temporal Directionality and Lag Controls

To test whether the transferred dynamics-sensitive subspace contains more than static state or smooth action information, we performed a separate episode-level leave-one-out audit. For each external window and horizon h, the target is the true encoded latent increment in a source-fitted basis band. The feature arms are a future-action summary aligned with that increment, a same-length action summary shifted back by one latent step, the current latent state, and the corresponding state-augmented action features. A within-episode derangement swaps the two action sources while preserving their marginal distribution. The primary contrast is

$$\Delta_{b,h}^{\text{future}-\text{past}} = R^2(\text{future action} + \text{state}) - R^2(\text{past action} + \text{state}), \tag{S7}$$

and a second contrast compares future-action R^2 with the deranged-action control. Inference uses episode-cluster bootstrap intervals; no external window is treated as an independent episode.

The transferred dynamics basis results are shown in Table 23. A cell is counted as supported only when both 95% lower bounds (future versus past and future versus deranged) are positive. None of the 36 cells across the two basis bands, three horizons, and three normalizations satisfies this strict criterion; 21 cells have positive point estimates for both contrasts, while 15 do not. The directionality audit therefore rules out a stronger normalization- invariant temporal-action claim on this external sample. It does not negate the separate held-out evidence that a dynamics-sensitive subspace can be extracted or that its coordinates affect frozen rollout error.

Table 23: Temporal directionality controls for the transferred dynamics basis. Entries are future-plus-state minus past-plus-state R^2, followed by future-action minus deranged-action R^2; intervals are 95% episode bootstrap intervals.

Normalization	Band	h	$\Delta^{\text{future}-\text{past}}$	$\Delta^{\text{future}-\text{deranged}}$
public	top-8	1	-0.0345 $[-0.1301, 0.0638]$	$+0.1465$ $[-0.0439, 0.2522]$
public	top-8	4	$+0.0084$ $[-0.0925, 0.1304]$	$+0.1687$ $[-0.0570, 0.4288]$
public	top-8	8	-0.0208 $[-0.1601, 0.1467]$	$+0.0855$ $[-0.0439, 0.2194]$
public	middle-8–32	1	$+0.4477$ $[+0.1260, +0.8129]$	$+0.1436$ $[-0.0189, +0.3490]$
public	middle-8–32	4	$+0.1050$ $[-0.1328, +0.3587]$	$+0.0864$ $[-0.0479, +0.2390]$
public	middle-8–32	8	$+0.1007$ $[-0.0848, +0.2776]$	$+0.1106$ $[-0.0033, +0.2432]$
validation z-score	top-8	1	-0.0345 $[-0.1300, 0.0638]$	$+0.1465$ $[-0.0439, +0.2522]$
validation z-score	top-8	4	$+0.0101$ $[-0.0936, +0.1358]$	$+0.1687$ $[-0.0570, +0.4287]$
validation z-score	top-8	8	-0.0206 $[-0.1606, +0.1482]$	$+0.0855$ $[-0.0439, +0.2194]$
validation z-score	middle-8–32	1	$+0.4475$ $[+0.1271, +0.8125]$	$+0.1436$ $[-0.0189, +0.3489]$
validation z-score	middle-8–32	4	$+0.0997$ $[-0.1615, +0.3762]$	$+0.0864$ $[-0.0479, +0.2389]$
validation z-score	middle-8–32	8	$+0.1087$ $[-0.0938, +0.2975]$	$+0.1106$ $[-0.0033, +0.2432]$
raw/100	top-8	1	-0.0342 $[-0.1290, 0.0635]$	$+0.1465$ $[-0.0439, +0.2522]$
raw/100	top-8	4	-0.0129 $[-0.0687, 0.0421]$	$+0.1687$ $[-0.0570, +0.4287]$
raw/100	top-8	8	-0.0239 $[-0.1291, 0.1063]$	$+0.0855$ $[-0.0439, +0.2194]$
raw/100	middle-8–32	1	$+0.4269$ $[+0.1310, +0.7684]$	$+0.1436$ $[-0.0189, +0.3489]$
raw/100	middle-8–32	4	$+0.0788$ $[-0.0036, +0.1601]$	$+0.0864$ $[-0.0479, +0.2390]$
raw/100	middle-8–32	8	$+0.0846$ $[-0.0145, +0.1845]$	$+0.1106$ $[-0.0033, +0.2432]$

This control is deliberately stronger than a nonzero-action versus zero-action comparison, but it remains a predictive temporal diagnostic. A positive future-minus-past contrast would indicate temporal precedence in the encoded readout, not a counterfactual action effect or physical controllability.

12.3 Pixel-to-Latent Intervention with a Learned Encoder

The frozen-adapter study leaves open whether the same boundary is visible when the representation itself is learned. We therefore repeat the intervention in a small action-conditioned pixel world. A convolutional encoder and latent predictor are trained jointly from 16×16 torus images using next-embedding prediction plus SIGReg. Horizontal actions translate by one pixel and vertical actions by two pixels; the one-step training split includes a fixed action-group-specific scene offset, while the rollout split uses random action sequences with exact torus ground truth. All conditions use the same generated splits, initialization seeds, optimizer, early-stopping rule, and evaluation code. The result is a learned-encoder diagnostic, not an end-to-end LeWorldModel or PushT experiment.

Table 24: Pixel-to-latent alignment intervention in a synthetic torus world. The encoder and action-conditioned predictor are trained jointly; entries are means over three seeds. One-step and $h = 8$ rollout MSE are held-out latent prediction errors, and composition error averages the 16 ordered two-action products. The two commutators are diagnostics of the learned latent geometry, not task-performance measures.

Mode	Weight	One-step MSE	$h = 8$ rollout MSE	Composition error	Global $\mathcal{C}$	Conditional $\mathcal{C}$	Stable rank
None	0	0.000253	0.000262	1.25651	0.183362	0.252623	2.00777
Global	0.01	0.000549	0.000555	1.62688	0.064230	0.108349	2.19314
Global	0.1	0.002047	0.002004	2.40440	0.038367	0.046918	3.42851
Global	1	0.012705	0.013520	1.96097	0.100343	0.084603	2.79232
Conditional	0.01	0.000393	0.000421	1.59206	0.019057	0.013691	1.89100
Conditional	0.1	0.001638	0.001729	1.93547	0.027033	0.046022	2.78120
Conditional	1	0.013387	0.014319	2.12862	0.030234	0.021923	2.44613

Table 24 reproduces the qualitative boundary from the linear benchmark after introducing a trainable pixel encoder. The conditional penalty at weight 0.01 reduces the conditional commutator from 0.252623 to 0.013691, while one-step MSE, rollout MSE, and composition error all increase relative to the unregularized model. Global alignment at weight 0.1 similarly lowers the pooled commutator to 0.038367 while increasing one-step MSE by roughly an order of magnitude and producing the largest composition error. The learned pixel benchmark therefore supports a more precise intervention claim:

alignment penalties can move conditional geometry in a trainable representation, but commutator reduction alone does not imply improved prediction, composition, or rollout stability. Because the world is synthetic and the encoder is intentionally small, this result does not support claims about nonlinear LeWorldModel training, PushT performance, online planning, causal controllability, or GPU efficiency.

12.4 Latent and Predictor Equivariance in a Controlled Vector World

The preceding experiments target covariance–predictor alignment without assuming a group action. To isolate a complementary question, we construct a controlled action-conditioned vector world whose transition is $s_{t+1} = s_t + a_t$. States and actions are four-dimensional vectors made of two two-dimensional rotation blocks. A 90° block rotation is applied jointly to state and action, and an eight-dimensional block rotation is used as the prescribed latent representation. This is a synthetic mechanism benchmark; it is not a LeWorldModel or PushT retraining experiment.

For an encoder E, predictor P, state transform T, action transform T_a, and latent representation ρ, the latent and predictor interventions are defined in Eqs. (S8) and (S9):

$$\mathcal{L}_{\text{latent}} = \|E(Ts) - \rho(E(s))\|_2^2 , \tag{S8}$$

and

$$\mathcal{L}_{\text{predictor}} = \|P(E(Ts), T_a a) - \rho(P(E(s), a))\|_2^2 . \tag{S9}$$

The four modes are no intervention, latent-only, predictor-only, and both. All modes use identical generated splits, initialization seeds, optimizer, validation-MSE checkpoint selection, and evaluation code. The objective contains the prediction loss and the declared intervention only; an unrelated representation regularizer is not included. We use three seeds and weights 1 and 10 for the intervention modes, and report sample standard deviations rather than inferential significance tests.

Table 25: Latent–predictor equivariance ablation in the controlled vector world. Entries are mean $\pm$ sample standard deviation over three seeds. Lower is better for all error columns; stable rank is a collapse diagnostic, not a performance objective. Checkpoints are selected by held-out one-step MSE, not by an equivariance metric.

Mode	Weight	One-step MSE	$h = 8$ rollout MSE	Composition error	Latent equiv. error	Predictor equiv. error	Stable rank
None	0	$1.17\times10^{-4} \pm 3.7\times10^{-6}$	$3.50\times10^{-4} \pm 3.7\times10^{-5}$	0.175 ± 0.024	1.414 ± 0.004	1.415 ± 0.002	4.581 ± 0.353
Latent	1	$9.97\times10^{-5} \pm 1.1\times10^{-5}$	$2.86\times10^{-4} \pm 4.2\times10^{-5}$	0.979 ± 0.010	1.289 ± 0.058	1.321 ± 0.058	4.837 ± 0.327
Latent	10	$3.97\times10^{-4} \pm 1.6\times10^{-5}$	$2.53\times10^{-3} \pm 2.8\times10^{-4}$	0.386 ± 0.077	0.204 ± 0.069	0.227 ± 0.049	2.689 ± 0.132
Predictor	1	$1.16\times10^{-4} \pm 1.3\times10^{-5}$	$3.03\times10^{-4} \pm 3.1\times10^{-5}$	0.998 ± 0.006	1.366 ± 0.046	1.349 ± 0.027	4.986 ± 0.370
Predictor	10	$1.90\times10^{-4} \pm 4.0\times10^{-5}$	$3.87\times10^{-4} \pm 6.5\times10^{-5}$	1.074 ± 0.051	1.367 ± 0.041	0.573 ± 0.126	4.879 ± 0.213
Both	1	$1.07\times10^{-4} \pm 1.3\times10^{-5}$	$2.77\times10^{-4} \pm 4.4\times10^{-5}$	0.998 ± 0.009	1.308 ± 0.055	1.310 ± 0.049	4.815 ± 0.386
Both	10	$8.50\times10^{-4} \pm 2.6\times10^{-4}$	$1.69\times10^{-3} \pm 7.6\times10^{-4}$	0.942 ± 0.045	0.433 ± 0.128	0.279 ± 0.057	3.328 ± 0.375

Table 25 separates the two intended interventions. At weight 10, latent-only training substantially reduces the latent equivariance error, whereas predictor-only training primarily reduces predictor equivariance; the combined intervention reduces both, though with a trade-off in one-step and rollout error. In particular, the baseline composition error is 0.175, while the strong predictor-only and combined interventions reach 1.074 and 0.942, respectively. Thus enforcing a symmetry diagnostic is not by itself evidence of improved composition or rollout. The result is complementary to the symmetry-agnostic alignment experiments above: it establishes separability of two equivariance interventions in a controlled vector world, not a universal learned-world-model law or a planning improvement.

12.5 OOD Rollout and Candidate-Set Planning Boundary

The preceding ablation identifies geometric effects, but it does not by itself test whether those effects transfer beyond the training distribution. We therefore reuse the same trained checkpoints and evaluate three distribution changes: doubling the action scale from 0.25 to 0.50, shifting the initial state by $+2$, and applying both changes together. We also use a controlled candidate-set planning proxy. For each task, one of 64 horizon-4 action sequences generates the goal; the model selects a candidate by predicted latent goal distance, and we score the selected physical goal distance against the oracle candidate. This is a synthetic planning proxy, not online task success.

Table 26: Functional consequences of equivariance interventions under OOD evaluation and a controlled candidate-set planning proxy. Entries are means over three seeds; the complete JSON also reports sample standard deviations and all per-seed records. OOD-action MSE uses twice the training action scale, and OOD-combined $h = 8$ MSE uses both the action-scale change and a $+2$ initial state shift. Planning success uses physical goal distance ≤ 0.25; planning regret is selected physical distance minus the oracle candidate distance.

Mode	Weight	ID one-step MSE	OOD-action MSE	OOD-combined $h = 8$ MSE	ID success	OOD success	ID regret	OOD regret
None	0	0.000117	0.000289	0.006022	0.0069	0.0174	1.381	2.828
Latent	1	0.000100	0.000254	0.006025	0.0174	0.0174	1.347	2.842
Latent	2	0.000120	0.000279	0.005909	0.0139	0.0278	1.299	2.789
Latent	3	0.000175	0.000343	0.006873	0.0347	0.0347	1.260	2.654
Latent	5	0.000265	0.000456	0.010925	0.0799	0.0521	1.074	2.337
Latent	10	0.000397	0.000610	0.014803	0.1840	0.1493	0.855	1.860
Predictor	1	0.000116	0.000253	0.005797	0.0139	0.0208	1.367	2.739
Predictor	2	0.000121	0.000251	0.005773	0.0139	0.0174	1.329	2.732
Predictor	3	0.000126	0.000258	0.005823	0.0139	0.0104	1.321	2.748
Predictor	5	0.000139	0.000286	0.005865	0.0139	0.0139	1.338	2.728
Predictor	10	0.000190	0.000413	0.006320	0.0313	0.0104	1.251	2.726
Both	1	0.000107	0.000240	0.005956	0.0069	0.0174	1.340	2.742
Both	2	0.000121	0.000249	0.005797	0.0278	0.0243	1.258	2.736
Both	3	0.000157	0.000296	0.005749	0.0347	0.0174	1.241	2.762
Both	5	0.000293	0.000469	0.006187	0.0278	0.0313	1.247	2.601
Both	10	0.000850	0.001079	0.009892	0.0347	0.0278	1.228	2.535

Table 26 shows a non-monotone functional trade-off. Predictor-only weights 1–3 keep ID one-step error close to the baseline and slightly lower OOD-action rollout error, but planning regret changes only modestly. Latent-only weights 5–10 reduce the candidate-set planning regret and raise proxy success, while substantially increasing one-step and long-horizon prediction error. The combined intervention follows a similar pattern. Thus this controlled study does not establish that lower equivariance error improves planning; it shows that geometry can change OOD and candidate-set behavior independently of a single one-step score, while a matched causal interpretation still requires explicit one-step matching.

13 Frozen I-JEPA Feature Measurements

The I-JEPA measurement repeats the covariance-relative readout in a second frozen predictive representation family. We use the official I-JEPA ViT-H/16 checkpoint, encode 30 balanced images each from six standard vision benchmarks, and compare each source image with a horizontal flip. The source features define the covariance eigenspace. The feature difference between the transformed and source image defines the displacement whose subspace and energy concentration are measured.

Across the six benchmarks, the random-subspace mean stays near 0.0124 and the norm-matched random lift stays near 1.0, while the observed overlap and lift vary by benchmark. Table 27 reports the complete measurements visualized in the main text.

Table 27: Frozen I-JEPA view-transfer readouts across standard benchmarks. Each row uses 30 balanced images and a horizontal-flip view change. Displacement overlap is compared with a random-subspace reference, and top-16 energy lift with norm-matched random directions.

Dataset	Images	Displacement overlap	Random-subspace overlap	Top-16 lift	Random-direction lift	Source effective rank	Displacement effective rank
CIFAR-10	30	0.21439	0.01237	29.98166	1.01503	12.41395	6.81819
CIFAR-100	30	0.16547	0.01259	21.15943	1.01217	15.06819	7.86112
STL10	30	0.11062	0.01245	12.00550	1.01267	17.59172	11.12838
SVHN	30	0.36572	0.01239	53.32585	1.03064	2.34480	3.32782
Tiny ImgNet	30	0.12657	0.01236	14.72093	1.00325	19.16375	9.45563
Food-101	30	0.17327	0.01229	30.01951	1.00183	18.40518	6.20927

To determine whether a displacement statistic also yields a stable estimated predictor operator, we fit a ridge linear map in source-PCA coordinates and evaluate it on held-out view pairs. The rank sweep covers CIFAR-10 with ranks 8 and 16 and scikit-learn-digits with ranks 8, 16, and 32. Held-out R^2 remains high (0.97141–0.99840), but the fitted covariance–operator commutator is below the spectrum-preserving random-basis mean in every fit, and the predicted-displacement lift is below the pair-shuffled contrast. Thus covariance-relative feature displacements can be reproducible while a

low-rank operator estimate remains sample- and rank-dependent. The two estimands require separate reporting in frozen JEPA representations.

Table 28: Held-out I-JEPA operator-rank sweep. A low-rank ridge map is fit from source to horizontal-flip features in source-PCA coordinates. The table reports held-out prediction and covariance-relative operator measurements across ranks.

Dataset	Rank	Held-out R^2	Normalized commutator $\mathcal{C}$	Random-basis mean $\mathcal{C}$	Predicted-displacement lift	Pair-shuffled lift
CIFAR-10	8	0.99840	0.00654	0.01071	0.45786	1.80179
CIFAR-10	16	0.99405	0.03366	0.05249	0.47367	1.83083
Digits	8	0.97141	0.03222	0.05201	0.49771	1.64230
Digits	16	0.97674	0.07861	0.17184	0.46940	1.85025
Digits	32	0.97609	0.16240	0.18092	1.63008	1.94565

The six-benchmark displacement measurements and the held-out operator fits therefore answer different questions. The former measures where a deterministic view change lies relative to source covariance; the latter additionally depends on the sample size, regression rank, and conditioning of an estimated linear map.

14 Held-out Temporal Measurements in Released World Models

The temporal experiments estimate action-conditioned linear operators from released JEPA-WM and DINO-WM checkpoints. Every evaluation reports latent MSE, the covariance–operator commutator, and a spectrum-preserving random-basis reference at horizons $h \in \{1, 2, 4\}$. The four checkpoint–environment combinations separate model family, environment, split provenance, and horizon while keeping the measurement vocabulary fixed.

14.1 JEPA-WM on PushT

We evaluate the public epoch-50 JEPA-WM PushT checkpoint on all 21 held-out validation trajectories. The protocol uses two encoded context frames, five simulator actions per conditioning block, and autoregressive horizons $h \in \{1, 2, 4\}$. The wrapper's missing historical action token is filled with the normalized mean action; this is a temporal-alignment correction and does not alter the released weights. The table reports latent prediction error and the covariance–operator commutator against a spectrum-preserving random-basis null for this released model and protocol.

Table 29: JEPA-WM on held-out PushT trajectories. Results use the public validation split (21 trajectories), two-frame context, and five-action conditioning blocks.

Prediction horizon h	Latent prediction MSE	Normalized commutator $\mathcal{C}$	Random-basis mean $\mathcal{C}$
1	0.07257	1.17128	1.39180
2	0.06465	1.59889	1.40656
4	0.09167	2.09273	1.41442

The commutator lies below the spectrum-preserving reference at horizon 1 and above it at horizons 2 and 4. Latent MSE does not follow the same ordering, showing that prediction error and covariance-relative alignment provide distinct information about this checkpoint.

14.2 DINO-WM on PointMaze

We apply the same latent-error and covariance–operator readouts to the public DINO-WM PointMaze release, providing an independent implementation and environment. The evaluation uses its epoch-10 predictor checkpoint, three context frames, five-step action blocks, and all 200 validation episodes from the released 2,000-episode archive. The public loader specifies a 0.9 split and implements episode-level "torch.randperm" splitting with seed 42; we reproduce those IDs from the released code and dataset ordering. The archive does not serialize the membership file, so episode membership is reconstructed from the released loader, seed, and dataset ordering. Inputs use the released "normalize_action=True" preprocessing, which computes action and proprio statistics over the full archive before the split.

Table 30: DINO-WM on PointMaze. The same latent-error and covariance–operator readouts are applied to a different action-conditioned world-model implementation and environment.

Prediction horizon h	Latent prediction MSE	Normalized commutator $\mathcal{C}$	Random-basis mean $\mathcal{C}$
1	0.00201	37.89996	1.46722
2	0.00491	61.50803	1.42252
4	0.01121	96.48006	1.42394

The observed commutator exceeds the spectrum-preserving reference at every horizon and increases with rollout length, while latent MSE also increases. This extends the readout to a second implementation and environment. Because the archive does not serialize episode membership, the released loader, seed, and dataset order define the evaluation protocol.

14.3 DINO-WM ON PUSHT

We evaluate the released DINO-WM PushT checkpoint on the public 18,685/21 train/validation directory split. The protocol uses the checkpoint's epoch-2 predictor, three context frames, five-step action blocks, and the same latent-error, commutator, and spectrum-preserving null readouts. Unlike the PointMaze evaluation, no episode membership is reconstructed: the validation directory itself is the held-out split.

Table 31: DINO-WM on the explicit PushT validation split. The temporal/operator readouts are evaluated on the released action-conditioned checkpoint with a serialized train/validation directory split.

Prediction horizon h	Latent prediction MSE	Normalized commutator $\mathcal{C}$	Random-basis mean $\mathcal{C}$
1	0.01672	0.62873	1.38166
2	0.02316	0.72850	1.38331
4	0.06510	2.08349	1.43895

The commutator is below the spectrum-preserving reference at horizons 1 and 2 and above it at horizon 4. Compared with the DINO-WM PointMaze result, this explicit-split PushT evaluation shows that the relation to the null is not a fixed property of the DINO-WM implementation; it changes with environment and horizon.

14.4 JEPA-WM ON POINTMAZE

For a matched-environment comparison, we evaluate the released JEPA-WM PointMaze checkpoint (epoch 50) on all 200 episodes in the validation side of the same public PointMaze archive using the released episode splitter (seed 42). The protocol uses two context frames and the same h=1/2/4 latent/operator readouts. We additionally report rank-truncated fits because a full 384-dimensional least-squares operator can be ill-conditioned even with 200 episodes.

Table 32: JEPA-WM on PointMaze. The checkpoint is evaluated on all 200 code-reconstructed validation episodes with the same readouts used for DINO-WM.

Prediction horizon h	Latent prediction MSE	Normalized commutator $\mathcal{C}$	Random-basis mean $\mathcal{C}$
1	4.98455	60.48199	1.45494
2	4.86646	43.44444	1.57804
4	4.70775	37.19464	1.53736

The full-rank fit is above the spectrum-preserving null at all three horizons, but its scale is sensitive to rank and sample conditioning. In rank-truncated fits (ranks 8, 16, 32, 64, 128), the observed commutator remains above the null at every tested rank and horizon; for example, at rank 8 the observed/null pairs are 1.97/1.10, 3.37/1.18, and 1.79/1.05 for $h = 1, 2, 4$. The persistence of this contrast after rank truncation indicates that the comparison is not specific to the full 384-dimensional fit, even though its absolute scale remains estimator-dependent.

14.5 Cross-Fitted Temporal Correspondence on PointMaze

To distinguish temporal correspondence from a high-dimensional same-sample fit, we fit rank-16 ridge operators on five training folds and score only held-out episodes. For JEPA-WM, held-out R^2 for the paired map is $0.082/0.183/0.025$ at $h = 1/2/4$, versus $-0.092/-0.094/-0.081$ after pair shuffling. On DINO-WM, the corresponding values are $0.935/0.808/0.554$ versus $-0.093/-0.083/-0.065$. A spectrum-preserving random-basis control gives JEPA-WM R^2 of $-0.471/-0.565/-0.327$ and DINO-WM values of $0.889/0.693/0.312$.

Table 33: Cross-fitted temporal-operator controls on PointMaze. Held-out means for rank 16 and ridge 10^{-2}. Paired and pair-shuffled fits use identical folds; the spectral control applies a random orthogonal basis rotation that preserves the fitted operator's singular spectrum.

Model	h	Paired held-out R^2	Pair-shuffled held-out R^2	Spectrum-preserving control R^2	Paired normalized commutator $\mathcal{C}$
JEPA-WM	1	0.082	-0.092	-0.471	2.572
JEPA-WM	2	0.183	-0.094	-0.565	3.623
JEPA-WM	4	0.025	-0.081	-0.327	2.070
DINO-WM	1	0.935	-0.093	0.889	0.404
DINO-WM	2	0.808	-0.083	0.693	0.612
DINO-WM	4	0.554	-0.065	0.312	0.972

The cross-fitted results distinguish linear correspondence from raw operator scale. DINO-WM is more linearly predictable in this PointMaze readout, whereas JEPA-WM shows a smaller paired signal that weakens at $h = 4$. The full-rank commutators remain useful as descriptive diagnostics, but the held-out fits are the relevant quantities for assessing correspondence under finite samples.

14.6 Action-Conditioning Interventions

An action-conditioned predictor can remain accurate for reasons other than the ordering of the supplied future actions. We therefore compare two derangements. A cross-episode derangement reassigns action blocks between episodes. A stricter within-episode derangement preserves the episode identity, target latents, and exact multiset of future five-step action blocks, changing only their temporal order. Each derangement is a zero-fixed-point bijection and is repeated with three permutation seeds. The permutation seeds change the intervention only; they do not create new independent trajectories.

Table 34: Action-order sensitivity of released checkpoints. Entries are mean changes in latent prediction MSE relative to observed actions across three derangement seeds. Positive values indicate worse prediction after the intervention. Cross-episode derangement changes episode/action pairing; within-episode derangement preserves the future-action block multiset and changes only its order.

Environment	Model	Prediction horizon h	Cross-episode ΔMSE	Within-episode ΔMSE
PushT	JEPA-WM	1	0.04691	0.04243
PushT	JEPA-WM	2	0.08668	0.05853
PushT	JEPA-WM	4	0.10367	0.08522
PushT	DINO-WM	1	0.04816	0.00000
PushT	DINO-WM	2	0.08970	0.03169
PushT	DINO-WM	4	0.13227	0.07507
PointMaze	JEPA-WM	1	0.00272	-0.00375
PointMaze	JEPA-WM	2	-0.01049	-0.01762
PointMaze	JEPA-WM	4	-0.03199	-0.03421
PointMaze	DINO-WM	1	0.01705	0.00000
PointMaze	DINO-WM	2	0.02390	0.01284
PointMaze	DINO-WM	4	0.02224	0.01667

PushT is consistently sensitive to future-action order for JEPA-WM and for DINO-WM beyond the first horizon. The DINO-WM horizon-1 within-episode value is exactly zero because its three context blocks are preserved and no shuffled future block has entered the one-step prediction. PointMaze separates the two checkpoints: DINO-WM remains positively sensitive at horizons 2 and 4, whereas

JEPA-WM does not deteriorate under the same intervention. This heterogeneity precludes interpreting action sensitivity as a universal property of either the architecture family or the commutator statistic. The intervention measures sensitivity of a fixed checkpoint to action order; it does not measure downstream control performance.

14.7 PointMaze Split Reconstruction Robustness

The public PointMaze archive does not serialize the episode-membership file. To test whether one code-reconstructed split drives the result, we evaluate both checkpoints on validation memberships generated by the released episode-level splitter with seeds 42, 0, and 1, and on an evaluator-defined tail holdout. The holdout overlap between these memberships is small (Jaccard 0.034–0.064). The ranges describe sensitivity of fixed checkpoints to membership choice; they are not independent training replicates.

Table 35: PointMaze robustness across reconstructed memberships. The operator fit uses rank 8 and ridge 10^{-2}. The final column is the held-out paired R^2 minus its pair-shuffled control; a positive range means paired temporal correspondence survives every evaluated membership.

Model	Prediction horizon h	Latent prediction MSE range	Paired-minus-shuffled held-out R^2 range
JEPA-WM	1	4.97558–4.99292	0.053–0.097
JEPA-WM	2	4.82610–4.87583	0.078–0.191
JEPA-WM	4	4.65493–4.70956	0.032–0.061
DINO-WM	1	0.00155–0.00201	0.977–1.024
DINO-WM	2	0.00386–0.00545	0.829–0.908
DINO-WM	4	0.00981–0.01121	0.600–0.656

The paired-minus-shuffled gap remains positive for both model families at all three horizons under every membership. Absolute full-rank commutators vary substantially across memberships—from 30.99 to 77.02 for JEPA-WM at $h = 1$, for example—so held-out operator controls are more informative than raw full-rank scale when comparing correspondence. The exercise reproduces the released splitting algorithm; without a serialized manifest, it cannot recover the membership used during the original training run.

14.8 Horizon, Context, Direction, and Sample-Size Ablations

We extend the horizon range to $h = 8$ using the same rank-8, ridge-10^{-2} held-out operator protocol. Paired R^2 remains above the pair-shuffled control for all four checkpoint–environment combinations: JEPA-WM PushT 0.277 versus -0.395, DINO-WM PushT 0.169 versus -0.425, JEPA-WM PointMaze -0.040 versus -0.044, and DINO-WM PointMaze 0.306 versus -0.044. At $h = 8$, the JEPA-WM PointMaze paired fit no longer shows a clear positive out-of-sample gap: it only narrowly exceeds the shuffled control. This result marks a limit of the chosen long-horizon estimator and is not inferred from the full-rank commutator alone.

The JEPA-WM external-context ablation compares one and two context frames while holding the internal predictor and preprocessing fixed. The effect is not monotone: one frame improves PushT horizon-1 MSE (0.02424 versus 0.03078), whereas two frames improve PushT horizon-2/4 MSE and PointMaze held-out operator identification at horizons 2 and 4. Reversing the fitted map from target-to-predicted to predicted-to-target leaves DINO-WM nearly unchanged but reduces the already-small JEPA-WM PointMaze paired R^2 from $0.059/0.133/0.024$ to $0.006/-0.004/-0.037$. Context length and operator direction are therefore analysis choices whose effects must be reported rather than treated as invariants.

Finally, nested PointMaze training subsets of 20, 40, 80, 120, and 160 episodes are evaluated against fixed 40-episode test sets over five split seeds. The mean paired-minus-shuffled gap remains positive at every size and horizon for both checkpoints. For JEPA-WM it ranges from 0.025 to 0.137 at $h = 1$, 0.147 to 0.205 at $h = 2$, and 0.048 to 0.152 at $h = 4$; the corresponding DINO-WM ranges are 0.955–1.260, 0.880–1.150, and 0.573–0.813. These measurements characterize estimator stability for frozen representations; they are not learning curves for model training.

15 FUNCTIONAL CONSEQUENCES OF ALIGNMENT INTERVENTIONS

The preceding alignment experiments establish that global and action-conditional penalties move different geometry diagnostics, but their repeated-action rollout does not test arbitrary action sequences. We therefore rerun the same learned action-conditioned linear benchmark with validation-MSE checkpoint selection and evaluate random group sequences through $h = 8$. We also use a controlled candidate-set planning proxy: an independent true target trajectory defines the goal, the model selects one of 64 candidates by predicted final-state distance, and physical regret is measured against the oracle candidate set. This remains a synthetic functional test rather than online task success.

Table 36: Functional consequences of global versus action-conditional alignment under random action-sequence rollout and a candidate-set planning proxy. The rollout horizon is $h = 8$; planning uses 96 tasks, 64 candidates, and horizon 4. Entries are means over three seeds; the full per-seed records and sample standard deviations are archived in the accompanying JSON artifact. Composition error is the mean normalized error over the four ordered two-action products. The global and conditional commutators are geometry diagnostics; planning regret and success are scored in the synthetic state space and are not online-control metrics.

Mode	Weight	ID one-step MSE	ID $h = 8$	OOD $h = 8$	Composition	ID regret	OOD regret	ID success	OOD success	Global $\mathcal{C}$	Conditional $\mathcal{C}$
None	0	2.44×10^{-6}	1.83×10^{-6}	2.51×10^{-6}	0.00830	0.000	0.000	1.000	0.983	0.356	3.30×10^{-5}
Global	0.01	2.14×10^{-5}	3.83×10^{-6}	2.40×10^{-6}	0.0146	0.000	0.000	1.000	0.983	0.352	5.59×10^{-5}
Global	0.1	0.00325	4.67×10^{-4}	7.48×10^{-4}	0.282	0.013	0.026	0.997	0.962	0.300	0.00186
Global	1	0.0770	27.302	225.747	2.733	0.316	0.358	0.618	0.403	0.0924	0.0373
Global	10	0.288	0.0740	0.939	1.919	0.580	0.447	0.490	0.413	0.00408	0.203
Global	100	0.350	0.0254	0.0160	1.384	0.620	0.384	0.434	0.417	0.000431	0.270
Conditional	0.01	2.23×10^{-6}	1.44×10^{-6}	2.46×10^{-6}	0.00774	0.000	0.000	1.000	0.983	0.356	2.72×10^{-5}
Conditional	0.1	2.40×10^{-6}	2.42×10^{-6}	2.48×10^{-6}	0.00796	0.000	0.000	1.000	0.983	0.356	3.26×10^{-5}
Conditional	1	8.27×10^{-6}	6.56×10^{-6}	2.96×10^{-6}	0.00914	0.000	0.000	1.000	0.983	0.356	4.59×10^{-5}
Conditional	10	0.00214	4.89×10^{-4}	8.39×10^{-4}	0.149	0.013	0.036	1.000	0.962	0.351	6.80×10^{-5}
Conditional	100	0.0576	0.0482	0.0155	1.519	0.263	0.368	0.649	0.451	0.265	1.71×10^{-4}

Table 36 gives a negative but informative functional result. Conditional alignment at weights 0.01–1 preserves the random-sequence rollout and candidate-set behavior of the unregularized model, whereas strong global alignment lowers the pooled commutator while increasing the conditional mismatch and sharply increasing composition and long-horizon error. Strong conditional alignment eventually exhibits its own weight-sensitive trade-off. Thus the experiment does not support a monotone claim that a lower commutator improves planning; it supports the narrower conclusion that global and conditional geometry changes have distinct functional consequences in this controlled synthetic world.

Exact action-sequence composition. The preceding functional benchmark evaluates random action sequences but reports composition only through ordered two-action products. We therefore retrain the same 11 mode/weight conditions with the same three seeds and enumerate every binary action sequence of length $h \in \{2, 4, 8\}$. For an ordered sequence $(a_1, \ldots, a_h)$, the structural product is $A_{a_h} \cdots A_{a_1}$; we compare the learned and data-generating products directly and do not assume that the two action labels form a closed group. The normalized Frobenius error is summarized by its mean, while the $h = 8$ P90 guards against an average being driven by only a subset of sequences.

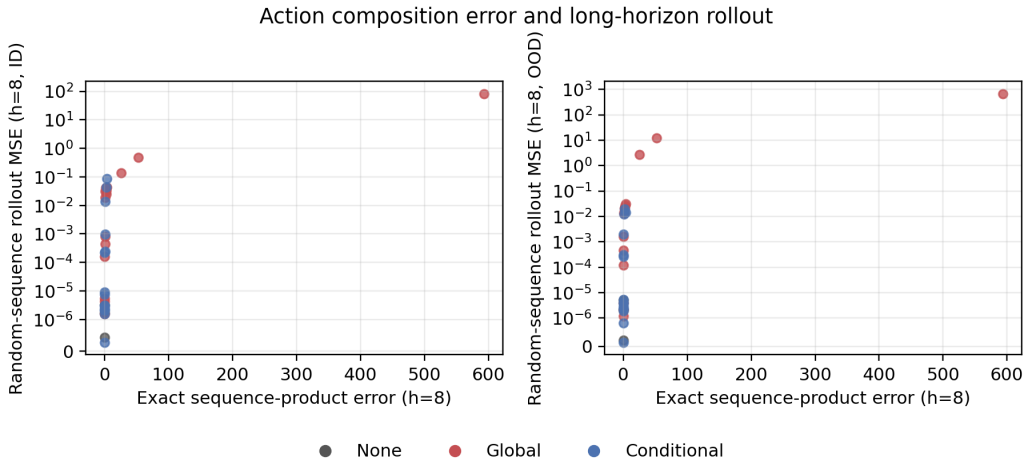
Table 37: Exact action-sequence composition and long-horizon rollout in the learned synthetic action-conditioned benchmark. Entries are means over three seeds; sample standard deviations are retained in the JSON artifact. Each composition column averages normalized Frobenius error over all binary action sequences of the indicated length (4, 16, and 256 sequences for $h = 2, 4, 8$); the P90 column is over the 256 length-eight products. Rollout errors use fresh random action sequences and are reported for the same horizon. This is a controlled linear diagnostic, not nonlinear LeWorldModel, PushT, or online planning evidence.

Mode	Weight	$E^{(2)}_{\mathrm{comp}}$	$E^{(4)}_{\mathrm{comp}}$	$E^{(8)}_{\mathrm{comp}}$	$E^{(8)}_{\mathrm{comp},90}$	ID rollout $h = 8$	OOD rollout $h = 8$
None	0	0.00830	0.00877	0.0127	0.0182	1.83×10^{-6}	2.51×10^{-6}
Global	0.01	0.0146	0.0147	0.0192	0.0254	3.83×10^{-6}	2.40×10^{-6}
Global	0.1	0.282	0.180	0.188	0.241	4.67×10^{-4}	7.48×10^{-4}
Global	1	2.733	10.562	215.968	325.854	27.302	225.747
Global	10	1.919	3.025	10.270	16.513	0.0740	0.939
Global	100	1.384	1.444	1.461	1.850	0.0254	0.0160
Conditional	0.01	0.00774	0.00779	0.0106	0.0161	1.44×10^{-6}	2.46×10^{-6}
Conditional	0.1	0.00796	0.00900	0.0136	0.0187	2.42×10^{-6}	2.48×10^{-6}
Conditional	1	0.00914	0.0121	0.0210	0.0287	6.56×10^{-6}	2.96×10^{-6}
Conditional	10	0.149	0.140	0.194	0.328	4.89×10^{-4}	8.39×10^{-4}
Conditional	100	1.519	1.764	2.166	3.819	0.0482	0.0155

Across the 33 per-seed records, the Pearson (Spearman) association between sequence-product error and same-horizon rollout error is 0.721 (0.920) at $h = 2$, 0.948 (0.949) at $h = 4$, and 0.996 (0.975) at $h = 8$ for ID sequences. The corresponding OOD values are 0.977 (0.971), 0.971 (0.943), and 0.997 (0.932). The $h = 8$ Pearson association remains positive within the global subset (ID 0.996, OOD 0.997) and the conditional subset (ID 0.987, OOD 0.892); a log-scale sensitivity analysis gives pooled values 0.984 and 0.991. These are descriptive associations rather than inferential or causal estimates.

Table 38: Association between exact sequence-product error and random-sequence rollout error across 33 per-seed records. Pearson and Spearman values are reported for ID and OOD sequences at each horizon. The table is descriptive and does not claim that composition error is a causal mediator of planning quality.

Horizon	ID Pearson	ID Spearman	OOD Pearson	OOD Spearman
2	0.721	0.920	0.977	0.971
4	0.948	0.949	0.971	0.943
8	0.996	0.975	0.997	0.932



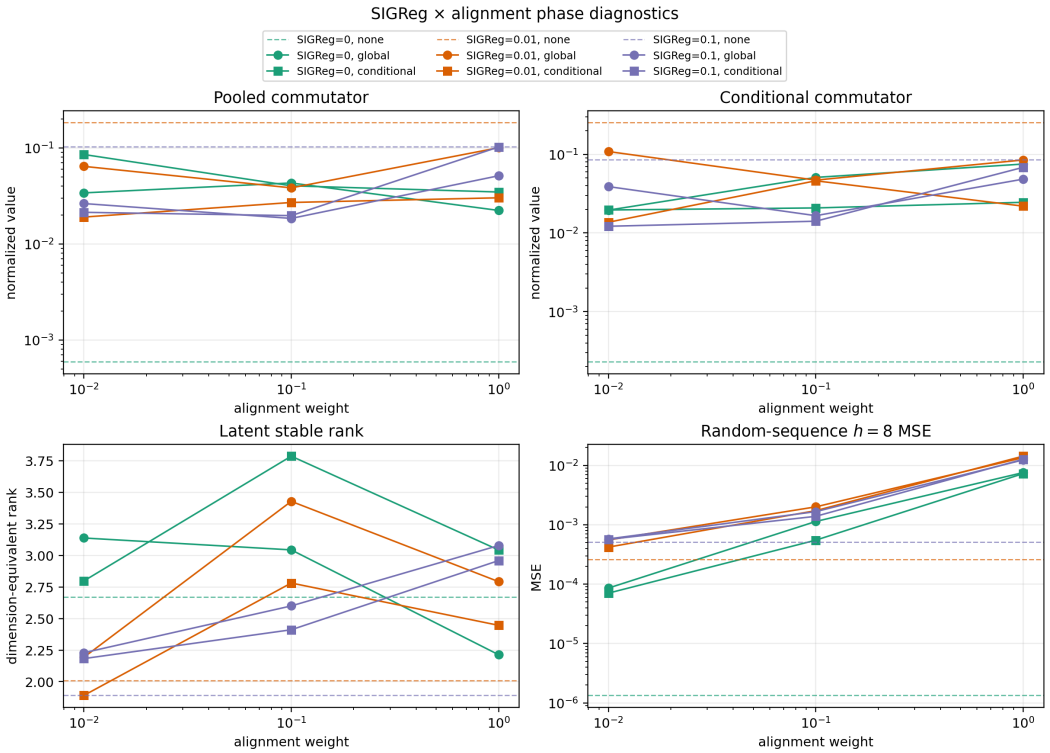
Supplementary Figure 2: Exact length-eight sequence-product error versus random-sequence rollout error for the ID and OOD evaluation splits. Colors identify the none, global, and conditional alignment interventions. The vertical axes use a symmetric-log scale to retain both near-zero and high-error conditions. The plot is a controlled synthetic association diagnostic, not evidence of causal control or online-planning improvement.

The sequence audit strengthens the functional interpretation of the benchmark: long-horizon error tracks the learned action-product structure more closely than a one-step or pooled commutator summary alone. It does not establish that alignment causes the association, that the relation transfers to nonlinear LeWorldModel dynamics, or that it predicts task-level planning success.

Separating SIGReg-induced isotropy from alignment. The preceding intervention still leaves a possible confound: SIGReg may change the latent covariance in a way that alters a covariance–predictor diagnostic without changing the predictor geometry in a useful sense. We therefore cross SIGReg weights $\{0, 0.01, 0.1\}$ with no alignment, global alignment, and action-conditional alignment at weights $\{0.01, 0.1, 1\}$. Each of the 21 conditions is evaluated with three matched seeds, identical generated splits, validation-MSE checkpoint selection, and the same learned pixel-world benchmark. In addition to the two commutators, we record the latent stable rank as an isotropy diagnostic and retain one-step, random-sequence $h = 8$, and ordered-composition readouts.

Table 39: SIGReg × alignment phase diagnostic in the learned synthetic pixel world. Entries are means over three seeds; per-seed values and sample standard deviations are in the JSON artifact. The no-alignment row is the matched baseline for each SIGReg weight. Stable rank is a covariance-shape diagnostic, not a measure of representation quality.

SIGReg	Mode	Align.	One-step	$h = 8$	Composition	Global $\mathcal{C}$	Conditional $\mathcal{C}$	Stable rank
0	None	0	1.219×10^{-6}	1.337×10^{-6}	0.0160	0.000593	0.000230	2.669
0	Global	0.01	8.650×10^{-5}	8.583×10^{-5}	0.1221	0.0340	0.0195	3.138
0	Global	0.1	1.107×10^{-3}	1.136×10^{-3}	0.3987	0.0431	0.0508	3.044
0	Global	1	7.232×10^{-3}	7.567×10^{-3}	0.7255	0.0223	0.0751	2.213
0	Conditional	0.01	6.792×10^{-5}	7.051×10^{-5}	0.1096	0.0853	0.0196	2.798
0	Conditional	0.1	5.231×10^{-4}	5.501×10^{-4}	0.2973	0.0403	0.0208	3.787
0	Conditional	1	6.729×10^{-3}	7.269×10^{-3}	0.8177	0.0347	0.0245	3.043
0.01	None	0	2.532×10^{-4}	2.616×10^{-4}	1.257	0.1834	0.2526	2.008
0.01	Global	0.01	5.489×10^{-4}	5.550×10^{-4}	1.627	0.0642	0.1084	2.193
0.01	Global	0.1	2.047×10^{-3}	2.004×10^{-3}	2.404	0.0384	0.0469	3.429
0.01	Global	1	1.271×10^{-2}	1.352×10^{-2}	1.961	0.1003	0.0846	2.792
0.01	Conditional	0.01	3.932×10^{-4}	4.205×10^{-4}	1.592	0.0191	0.0137	1.891
0.01	Conditional	0.1	1.638×10^{-3}	1.729×10^{-3}	1.935	0.0270	0.0460	2.781
0.01	Conditional	1	1.339×10^{-2}	1.432×10^{-2}	2.129	0.0302	0.0219	2.446
0.1	None	0	4.837×10^{-4}	5.068×10^{-4}	1.352	0.1031	0.0856	1.893
0.1	Global	0.01	5.580×10^{-4}	5.779×10^{-4}	1.548	0.0263	0.0388	2.229
0.1	Global	0.1	1.608×10^{-3}	1.665×10^{-3}	2.282	0.0185	0.0166	2.600
0.1	Global	1	1.236×10^{-2}	1.236×10^{-2}	2.834	0.0513	0.0483	3.078
0.1	Conditional	0.01	5.415×10^{-4}	5.683×10^{-4}	1.603	0.0214	0.0122	2.183
0.1	Conditional	0.1	1.348×10^{-3}	1.387×10^{-3}	1.813	0.0197	0.0141	2.411
0.1	Conditional	1	1.214×10^{-2}	1.268×10^{-2}	2.577	0.1021	0.0682	2.959



Supplementary Figure 3: Phase diagnostics for the SIGReg–alignment cross-factorial sweep. Dashed lines are the no-alignment baselines at each SIGReg weight; circles and squares denote global and conditional alignment, respectively. Lower commutators coexist with higher one-step, rollout, and composition errors in many $\mathrm{SIGReg} > 0$ conditions, so the diagnostic is not a sufficient functional objective.

The phase table makes the confound explicit. At $\mathrm{SIGReg} = 0$, adding alignment generally raises both commutators and worsens the functional readouts. At $\mathrm{SIGReg} = 0.01$ and 0.1, alignment often lowers the commutators relative to the matched no-alignment baseline, while every aligned condition has a higher mean $h = 8$ error and composition error in this sweep. Stable rank also moves with the regularizer and is non-monotone across alignment weights, confirming that covariance shape and functional prediction are separate estimands. This is CPU-only synthetic mechanism evidence: it rules out treating commutator reduction as sufficient evidence of better prediction in this setting, but it does not identify a universal causal effect or establish a nonlinear LeWorldModel, PushT, online-control, or GPU result.

15.1 Cross-source latent-dynamics transport in LeWorldModel

The preceding frozen PushT analyses identify a dynamics-sensitive latent subspace, but that identification alone does not show that the subspace is transported by autoregressive prediction. We therefore performed a separate cross-source audit. The covariance basis, raw dynamics cross-covariance basis, and action-residualized dynamics basis were fitted on an internal frozen latent export and transferred without refitting to the public external PushT validation trajectories. The external evaluation used 21 episodes and 164 windows (up to eight windows per episode), with leave-one-episode-out fits and episode-cluster bootstrap intervals.

For a basis B and coordinate band I, let $z_t^{B,I}$ denote the selected coordinates of the centered latent. We distinguish the encoded future increment from the autoregressive increment by

$$\Delta_h^{B,I} = z_{t+h}^{B,I} - z_t^{B,I}, \qquad \widehat{\Delta}_h^{B,I} = \widehat{z}_{t+h}^{B,I} - z_t^{B,I}, \tag{S10}$$

where the second quantity uses the frozen predictor and the same observed action sequence. A probe fitted on encoded trajectories was applied to both quantities. The resulting predicted-R^2 contrast compares the transported increment in the dynamics basis with a covariance or dimension-matched random top-8 basis on the same held-out windows. This is deliberately a latent transport estimand: it is not a physical-state decoder and does not intervene on the environment.

Table 40: Paired predicted-R^2 contrasts for the transferred dynamics top-8 basis. Each entry is dynamics basis minus the indicated control, with a 95% episode-cluster bootstrap interval. The basis is fit outside the external validation trajectories.

Action normalization	Horizon	Versus covariance	Versus random
Public mean/std	$h = 1$	$+0.0678$ $[0.0403, 0.0973]$	$+0.0622$ $[0.0330, 0.0989]$
Public mean/std	$h = 4$	$+0.0908$ $[0.0297, 0.1626]$	$+0.0460$ $[0.0063, 0.0927]$
Public mean/std	$h = 8$	$+0.1788$ $[0.0619, 0.3462]$	$+0.0670$ $[0.0142, 0.1240]$
Validation z-score	$h = 1$	$+0.0546$ $[-0.0130, 0.1215]$	$+0.0203$ $[-0.0372, 0.0808]$
Validation z-score	$h = 4$	$+0.4272$ $[0.1865, 0.7096]$	$+0.1384$ $[0.0118, 0.2933]$
Validation z-score	$h = 8$	$+0.5373$ $[0.3494, 0.7201]$	$+0.1470$ $[0.0187, 0.2797]$
Raw action$/100$	$h = 1$	-0.0128 $[-0.0402, 0.0106]$	$+0.0133$ $[-0.0232, 0.0496]$
Raw action$/100$	$h = 4$	$+0.0628$ $[-0.0675, 0.2124]$	-0.0487 $[-0.1332, 0.0225]$
Raw action$/100$	$h = 8$	$+0.0328$ $[-0.1555, 0.1820]$	-0.1485 $[-0.4477, 0.0396]$

Under the public normalization, the dynamics basis exceeds both controls at all three horizons. Under validation-split z-scoring, the advantage is supported at $h = 4$ and $h = 8$ but not at $h = 1$. Under raw action scaling, all paired intervals include zero. The transported dynamics top-8 predicted-R^2 in the public arm is 0.9064, 0.9057, and 0.7134 at $h = 1, 4, 8$ respectively; the decline at $h = 8$ is evidence against a claim of perfect long-horizon transport. The cross-normalization result therefore supports a conditional latent-transport statement, not a normalization-invariant dynamics discovery.

The control hierarchy is important for interpretation. Action-only and current-state-only probes are retained as predictive baselines, while the rollout-increment probe tests whether the frozen autoregressive trajectory preserves the transferred coordinates. The same external data also show that action summaries explain most physical displacement in the available probe; therefore the latent-transport result must not be relabeled as additional physical-motion prediction, causal action effect, online planning, task success, or retraining improvement. The complete machine-readable reports, including the three normalization arms and source hashes, are included with the experiment artifacts.

15.2 Cross-horizon latent-displacement transport

The preceding transfer test evaluates a fitted predictor readout at each horizon. We add a complementary test that does not fit any readout on the external trajectories. For each external window, let z_h denote the encoded future LeWM embedding at horizon h. For a source-fitted orthonormal basis B_k, we measure the fraction of the encoded latent displacement captured by its first k coordinates:

$$\rho_{B_k}(h_a, h_b) = \frac{\left\| (z_{h_b} - z_{h_a})^\top B_k \right\|_2^2}{\| z_{h_b} - z_{h_a} \|_2^2 + \epsilon}, \qquad (h_a, h_b) \in \{(1,2), (2,4), (4,8)\}. \tag{S11}$$

Here B_k is fitted only on the internal frozen latent export and is then held fixed. We compare the dynamics cross-covariance basis with the covariance basis and dimension-matched random subspaces; the conditional dynamics basis is reported as an additional sensitivity control. The external sample has 21 episodes and 42 windows. Inference is clustered by episode, and the three source fits are retained as separate estimates rather than pooled into a new interval.

Table 41: Cross-horizon transport of a source-fitted dynamics basis. Each row reports the mean top-8 projection-energy share across three independent source fits, followed by paired differences against the covariance and random subspace controls. Positive differences favor the dynamics basis. The source fit estimates are not pooled into a confidence interval; all three source fits have the displayed sign.

Encoded displacement	Dynamics share	Δ vs covariance	Δ vs random	Covariance sign	Random sign
$z_2 - z_1$	0.176089	$+0.105223$	$+0.134697$	All positive	All positive
$z_4 - z_2$	0.160430	$+0.095658$	$+0.119019$	All positive	All positive
$z_8 - z_4$	0.120726	$+0.050672$	$+0.079402$	All positive	All positive

The same conclusion holds for the wider top-32 band: the dynamics-minus- covariance contrasts are $+0.179237$, $+0.173664$, and $+0.074898$ for the three displacement spans, respectively, and every source fit remains positive. This is stronger than the earlier source-domain readout because the external trajectories are used only as held-out targets, not to define the basis. It supports a conditional statement that the source-fitted dynamics geometry captures external encoded latent evolution across multiple spans. It does not establish that the coordinates represent extra physical motion: the target in Eq. (S11) is an embedding displacement, while the separate action-conditioned physical probe does not show incremental physical-motion predictability beyond observed actions. Nor does this test establish normalization-invariant transport, causal controllability, online planning, task success, or retraining improvement.

15.3 Action-conditioned residual transport

To make the action-related alternative explicit, we next remove the prediction of a compact action-history probe before measuring the cross-horizon geometry. For an external window from episode e, let $A^e_{a:b}$ contain the mean action in each five-frame action block between horizons a and b. The probe $\widehat{f}_{-e}$ is fitted on all episodes except e, and the residual target is

$$\widetilde{\Delta}^e_{a:b} = (z^e_b - z^e_a) - \widehat{f}_{-e}(A^e_{a:b})\,. \tag{S12}$$

The feature vector concatenates the mean, first, and last action-block means; this is the same compact action probe used by the physical-motion audit. We use a ridge readout and evaluate it leave-one-episode-out. Thus, the action residual for a held-out episode is not computed by a model trained on that episode.

Table 42: Action-conditioned residual transport under the public action normalization. The action R^2 is cross-fitted against encoded latent displacement; negative values indicate that this compact action summary does not predict the target better than the training-centroid baseline. The final two columns are top-8 dynamics-basis projection contrasts averaged over three source fits.

Displacement	Action R^2	Residual/raw energy	Δ vs covariance	Δ vs random
$z_2 - z_1$	-0.071033	1.044834	$+0.105807$	$+0.117693$
$z_4 - z_2$	-0.176546	1.144078	$+0.090278$	$+0.104822$
$z_8 - z_4$	-0.226125	1.186186	$+0.046750$	$+0.073818$

The positive contrasts persist for every source fit and for all three action normalization arms (public, validation-split z-score, and raw action divided by 100). However, the negative action-probe R^2 and residual/raw energy above one are important diagnostics: the compact probe is weak for predicting this latent target, so Eq. (S12) must not be interpreted as removing all action effects. The defensible conclusion is narrower: the dynamics-basis enrichment is stable after the declared cross-fitted action control in the encoded latent space. It still does not establish additional physical-motion information, causal controllability, online planning, task success, or retraining gains.

15.4 Action-conditioned operator algebra

The residual analysis above controls for an observed action summary but does not test whether actions enter the latent transition in a compositional form. We therefore fit an explicit low-dimensional action-conditioned map in each leave-one-episode-out fold. Let ξ_t be the rank-16 source-PCA coordinate of either the encoded target trajectory or the autoregressively predicted trajectory, and let u_t be the mean

of the normalized five-frame action block aligned with the transition $\xi_t \to \xi_{t+1}$. The fitted map is

$$T_u(\xi) = A\xi + Bu + b, \tag{S13}$$

with a state-only map $T(\xi) = A_0\xi + b_0$ as the matched control. For the primary audit, all maps are fitted on the other twenty episodes and scored only on the held-out episode; the two windows per episode are never split across folds. We additionally repeat the public-normalization audit with up to eight windows per episode, retaining all available windows for short episodes.

We evaluate action alignment by replacing u_t with zero or with a cyclic permutation from another held-out transition. We evaluate composition by comparing repeated one-step application with a separately fitted direct map $D_H(\xi_t, u_t, \ldots, u_{t+H-1})$:

$$\delta_H = \frac{\left\| T_{u_{t+H-1}} \circ \cdots \circ T_{u_t}(\xi_t) - D_H(\xi_t, u_t, \ldots, u_{t+H-1}) \right\|_2}{\left\| D_H(\xi_t, u_t, \ldots, u_{t+H-1}) \right\|_2 + \epsilon}. \tag{S14}$$

Finally, we report a negated-action return proxy,

$$\iota(\xi, u) = \frac{\left\| T_{-u}(T_u(\xi)) - \xi \right\|_2}{\left\| \xi \right\|_2 + \epsilon}, \tag{S15}$$

which is an algebraic diagnostic rather than an assumption that contact dynamics are physically reversible.

Table 43: Cross-fitted action-alignment diagnostics on the external PushT validation trajectories. Entries are means with 95% episode-bootstrap intervals. "Action gain" is state-only MSE minus action-conditioned MSE; "permuted penalty" is permuted-action MSE minus observed-action MSE.

Trajectory source	State-only MSE	Action-conditioned MSE	Action gain	Zero penalty	Permuted penalty
Encoded target	0.6492	0.6525	-0.0033 $[-0.0137, 0.0071]$	-0.0033 $[-0.0148, 0.0077]$	$+0.0158$ $[0.0084, 0.0248]$
Autoregressive prediction	0.6532	0.6546	-0.0014 $[-0.0117, 0.0097]$	-0.0021 $[-0.0123, 0.0085]$	$+0.0147$ $[0.0078, 0.0222]$

The action permutation penalty is positive for both trajectory sources, while the independent one-step action gain and zero-action penalty are inconclusive. Thus, the stronger supported statement is that the fitted latent transition is sensitive to action–state alignment, not that a compact linear action term provides an independent one-step predictor improvement.

Table 44: Action composition and inverse-proxy diagnostics. Composition defect is defined in Eq. (S14); the final column is the mean Eq. (S15). The composed-minus-direct MSE contrasts use the same held-out windows.

Trajectory source	Horizon	Composition defect	Composed MSE	Direct MSE	Composed − direct MSE
Encoded target	2	0.165	1.679	1.682	-0.003 $[-0.021, 0.017]$
Encoded target	4	0.583	3.334	3.349	-0.016 $[-0.136, 0.104]$
Encoded target	8	1.015	5.163	9.280	-4.118 $[-5.709, -2.726]$
Autoregressive prediction	2	0.169	1.655	1.666	-0.011 $[-0.028, 0.006]$
Autoregressive prediction	4	0.565	3.127	3.246	-0.119 $[-0.275, 0.035]$
Autoregressive prediction	8	0.972	4.042	7.559	-3.517 $[-4.829, -2.349]$

Composition defect grows with horizon, but the composed-minus-direct contrast is inconclusive at $H = 2$ and $H = 4$ and reverses at $H = 8$ because the direct high-dimensional map is itself poorly identified. The negated-action return defect is 0.372 for encoded targets and 0.350 for predictions, and the episode-level correlation between $H = 8$ composition defect and observed $H = 8$ rollout MSE is weak (Pearson -0.115 and 0.228, respectively). Accordingly, these results support action-order sensitivity and expose a non-compositionality diagnostic, but do not support a claim of physical inverse consistency, causal controllability, or a monotone composition–rollout law.

To test whether this conclusion depends on using only two windows per episode, we repeated the public-normalization audit with up to eight windows per episode. The replication contains 164 windows from the same 21 episodes; the two shortest episodes contribute six windows, and all other episodes contribute eight. The folds, rank-16 PCA construction, ridge parameter, and episode bootstrap are unchanged. The replication is summarized in Tables 45 and 46.

Table 45: Eight-window replication of action-alignment diagnostics under the public action normalization. Entries are means with 95% episode-bootstrap intervals. The replication preserves the positive action-permutation penalty, whereas the independent one-step action gain remains inconclusive.

Trajectory source	Action gain	Zero penalty	Permuted penalty
Encoded target	$+0.0082\ [-0.0048, 0.0214]$	$+0.0078\ [-0.0057, 0.0224]$	$+0.0254\ [0.0144, 0.0400]$
Autoregressive prediction	$+0.0083\ [-0.0034, 0.0202]$	$+0.0077\ [-0.0048, 0.0207]$	$+0.0242\ [0.0148, 0.0359]$

Table 46: Eight-window replication of composition diagnostics. Entries are means with 95% episode-bootstrap intervals. δ_H is the repeated one-step versus direct-map defect; Δ_H is composed-minus-direct MSE.

Trajectory source	δ_2	δ_4	δ_8	Δ_2	Δ_4	Δ_8
Encoded target	$0.285\ [0.239, 0.337]$	$0.608\ [0.526, 0.708]$	$0.883\ [0.834, 0.929]$	$+0.009\ [-0.005, 0.023]$	$+0.048\ [-0.026, 0.121]$	$-0.293\ [-0.567, -0.047]$
Autoregressive prediction	$0.295\ [0.250, 0.344]$	$0.647\ [0.574, 0.739]$	$0.893\ [0.839, 0.953]$	$+0.006\ [-0.007, 0.019]$	$+0.061\ [-0.012, 0.138]$	$-0.189\ [-0.326, -0.051]$

The replication therefore strengthens the protocol-level observation that correct action–state pairing matters, but it does not promote the result to a composition law. The defect increases from $H = 2$ to $H = 8$ in both trajectory sources; Δ_H is compatible with zero at $H = 2$ and $H = 4$, then becomes negative at $H = 8$ because the separately fitted direct map is poorly identified at that horizon. This non-monotone contrast is precisely why we do not interpret the audit as evidence for an exact semigroup, a stable physical inverse, or online planning.

As a scale-robustness check, we repeated the complete frozen audit with validation-split z-scoring and with raw actions divided by 100. The action permutation penalty remains positive for encoded targets under all three normalizations and for predicted trajectories under the public and validation-split normalizations; the raw$/100$ predicted-trajectory interval includes zero. The state-only versus action-conditioned gain changes sign or becomes very small across arms, and the composition and inverse-proxy values also move with the ridge scale. We therefore treat the public-normalization table above as the primary arm inherited from the external temporal protocol and the following values as a robustness audit, not as pooled evidence.

Table 47: Normalization robustness for the external action-algebra audit. Entries are means; the primary public-normalization arm reports bootstrap intervals in Tables 43 and 44.

Normalization	Trajectory source	Action gain	Permuted penalty	$H = 8$ composition defect
Public mean/std	Encoded target	-0.0033	$+0.0158$	1.015
Public mean/std	Autoregressive prediction	-0.0014	$+0.0147$	0.972
Validation z-score	Encoded target	$+0.0010$	$+0.0244$	1.032
Validation z-score	Autoregressive prediction	$+0.0225$	$+0.0422$	0.916
Raw action$/100$	Encoded target	$+0.0038$	$+0.0042$	0.877
Raw action$/100$	Autoregressive prediction	$+0.0001$	$+0.0001$	0.203

This sensitivity check rules out a normalization-invariant claim that the compact linear action term always improves one-step prediction. It preserves a more limited conclusion: action-order perturbations can be detected in the latent operator, but their magnitude and the apparent long-horizon algebra depend on action scaling and regularization.

15.5 Conditional local transition geometry

The action-algebra audit establishes action-order sensitivity, but it does not directly test whether different action regimes correspond to different local transition maps. We therefore add a matched local-operator diagnostic on the same frozen external validation protocol. For each leave-one-episode-out fold, we fit a rank-16 PCA coordinate system on the remaining episodes and fit a ridge row operator separately within two action bins. The primary condition uses quantiles of the normalized five-frame action-block magnitude. A direction condition uses quantiles of the cosine-like projection onto a training-fold action axis; it tests whether the result is specific to direction rather than only movement amplitude. The random partition preserves the two bin sizes while independently permuting the labels in the training and held-out folds. As a stronger control, the state-matched partition permutes labels only within coarse strata of the first training-fold state-PCA coordinate. It therefore preserves

the condition-label composition within approximate state regions while breaking action–transition alignment.

Let $A_1, \ldots, A_m$ denote the fitted row transition operators for the action-aligned bins. We measure local transition-map heterogeneity by

$$\begin{aligned} \mathcal{H}(A_{1:m}) &= \frac{2}{m(m-1)} \sum_{i<j} \frac{\|A_i - A_j\|_F}{\|A_i\|_F + \|A_j\|_F + \epsilon}, \\ \Delta_{\mathcal{H}}^{\text{rand}} &= \mathcal{H}_{\text{aligned}} - \mathcal{H}_{\text{random}}, \\ \Delta_{\mathcal{H}}^{\text{state}} &= \mathcal{H}_{\text{aligned}} - \mathcal{H}_{\text{state-matched}}. \end{aligned} \tag{S16}$$

The estimand in Eq. (S16) is a descriptive difference between fold-fitted linear maps. It is not a nonlinear Jacobian estimate and does not assume that the action partition is a causal intervention.

Table 48: Conditional local transition geometry on the public PushT validation set. Each row reports mean differences between the action-aligned partition and the global random or state-matched null, with 95% episode-bootstrap intervals. $\Delta\mathcal{H}$ is the local operator heterogeneity from Eq. (S16); ΔR^2 is the state-matched held-out transition-prediction contrast. All operators are fit outside the held-out episode.

Normalization	Trajectory source / condition	$\Delta\mathcal{H}$ vs random	$\Delta\mathcal{H}$ vs state	ΔR^2 vs state
Public mean/std	Encoded / magnitude	$+0.0515$ $[0.0478, 0.0552]$	$+0.0518$ $[0.0487, 0.0550]$	-0.0046 $[-0.0297, 0.0170]$
Public mean/std	Encoded / direction	$+0.0456$ $[0.0416, 0.0499]$	$+0.0467$ $[0.0430, 0.0505]$	-0.0431 $[-0.1228, 0.0283]$
Public mean/std	Predicted / magnitude	$+0.0613$ $[0.0575, 0.0649]$	$+0.0609$ $[0.0564, 0.0658]$	$+0.0101$ $[-0.0416, 0.0487]$
Public mean/std	Predicted / direction	$+0.0558$ $[0.0517, 0.0599]$	$+0.0509$ $[0.0468, 0.0546]$	-0.0365 $[-0.0984, 0.0133]$
Validation z-score	Encoded / magnitude	$+0.0517$ $[0.0488, 0.0546]$	$+0.0504$ $[0.0465, 0.0547]$	$+0.0273$ $[-0.0002, 0.0596]$
Validation z-score	Encoded / direction	$+0.0418$ $[0.0381, 0.0455]$	$+0.0416$ $[0.0386, 0.0447]$	-0.0706 $[-0.1460, -0.0034]$
Validation z-score	Predicted / magnitude	$+0.0375$ $[0.0345, 0.0406]$	$+0.0405$ $[0.0367, 0.0444]$	-0.0072 $[-0.0324, 0.0167]$
Validation z-score	Predicted / direction	$+0.0440$ $[0.0398, 0.0484]$	$+0.0437$ $[0.0403, 0.0468]$	-0.0487 $[-0.1218, 0.0080]$
Raw action/100	Encoded / magnitude	$+0.0409$ $[0.0373, 0.0445]$	$+0.0412$ $[0.0375, 0.0451]$	-0.0099 $[-0.0257, 0.0057]$
Raw action/100	Encoded / direction	$+0.0437$ $[0.0399, 0.0484]$	$+0.0439$ $[0.0398, 0.0482]$	-0.0502 $[-0.1157, -0.0042]$
Raw action/100	Predicted / magnitude	$+0.0093$ $[0.0083, 0.0103]$	$+0.0094$ $[0.0086, 0.0102]$	$+0.0013$ $[-0.0410, 0.0406]$
Raw action/100	Predicted / direction	$+0.0111$ $[0.0101, 0.0121]$	$+0.0117$ $[0.0108, 0.0127]$	-0.0097 $[-0.0254, 0.0008]$

The local heterogeneity contrast against the state-matched null is positive with a positive bootstrap lower bound in all twelve source–condition– normalization cells. This is the strongest conclusion licensed by the diagnostic: action-aligned partitions reveal latent transition-map heterogeneity beyond both a bin-size null and a coarse state-only null, and the effect survives the direction control and all three action normalizations. The magnitude varies substantially, especially for the predicted trajectory under raw action scaling. The held-out R^2 contrast is not uniformly positive; direction conditioning is negative in all three normalization arms. For an additional granularity sensitivity check, we repeat the public-normalization state-matched permutation with 4, 8, and 16 first-coordinate strata. The resulting local-heterogeneity contrasts are shown in Table 49; every source–condition combination retains a positive lower confidence bound at all three stratification resolutions, with 16 test rows covered per held-out fold. Thus, detectable action-dependent regime structure must not be reported as a prediction gain, a causal action effect, an online-planning improvement, or evidence that an alignment regularizer would improve the trained model.

Table 49: Sensitivity of the state-matched null to coarse state-stratum count on the public PushT validation set. Each cell reports the mean $\Delta\mathcal{H}^{\text{state}}$ and its 95% episode-bootstrap interval for 4, 8, or 16 first-coordinate strata. All reports use the same 21 held-out episodes and 42 windows, with 2,000 episode-cluster bootstrap replicates per cell. The check concerns the granularity of a descriptive null, not a claim that any stratum count identifies a physical state partition.

Trajectory source	Condition	4 strata	8 strata	16 strata
Encoded	Magnitude	$+0.05107$ $[0.04723, 0.05477]$	$+0.05178$ $[0.04868, 0.05496]$	$+0.05247$ $[0.04861, 0.05622]$
Encoded	Direction	$+0.04353$ $[0.03975, 0.04762]$	$+0.04674$ $[0.04297, 0.05049]$	$+0.04673$ $[0.04327, 0.05017]$
Predicted	Magnitude	$+0.06232$ $[0.05970, 0.06491]$	$+0.06091$ $[0.05639, 0.06584]$	$+0.05930$ $[0.05483, 0.06420]$
Predicted	Direction	$+0.05431$ $[0.05081, 0.05798]$	$+0.05091$ $[0.04681, 0.05462]$	$+0.05208$ $[0.04776, 0.05587]$

15.6 Leakage-controlled forward readout of latent dynamics

The projection-share and rollout-intervention measurements above show that a source-fitted dynamics basis is structurally and functionally involved in a frozen latent rollout. A stricter test asks whether

the same information can be read out *forward*: the future target must not be supplied as an input. We therefore use the released public PushT validation episodes and fit ridge probes in a leave-one-episode-out protocol. For each window, the features are the current latent coordinates in a source-fitted basis band, future action summaries, or their concatenation. The target is the future increment projected into the fixed top-eight coordinates of the source-fitted dynamics basis. Past-action and globally deranged-action features are temporal controls. All source bases are fit on a separate frozen latent export; no basis is refit on the external episodes.

Table 50: Leakage-controlled forward readout of a fixed dynamics target at $h = 8$. Each row uses the same target, 21 held-out episodes, eight windows per episode when available, and 2,000 episode-bootstrap replicates. R^2 is computed on the held-out episode using the training-fold target mean; the gain is the paired state-plus-action minus action-only contrast. The final column compares each input basis with the dynamics top-8 input on the same target.

Input basis (top-8)	Action R^2	State+action R^2	Gain over action	Dynamics minus input
Covariance	$+0.016$ $[-0.044, +0.083]$	-0.077 $[-0.182, +0.011]$	-0.093 $[-0.183, -0.010]$	$+0.562$ $[0.474, 0.678]$
Dynamics	$+0.016$ $[-0.044, +0.083]$	$+0.485$ $[0.423, +0.544]$	$+0.469$ $[0.401, +0.536]$	–
Residual	$+0.016$ $[-0.044, +0.083]$	-0.040 $[-0.119, +0.034]$	-0.056 $[-0.116, +0.004]$	$+0.525$ $[0.456, 0.607]$
Random	$+0.016$ $[-0.044, +0.083]$	-0.087 $[-0.168, -0.012]$	-0.103 $[-0.151, -0.058]$	$+0.572$ $[0.508, 0.643]$

The specificity is not an artifact of one action preprocessing or one ridge choice. With the matched-band target, the dynamics top-8 state-plus-action gain is $+0.469$ $[0.401, 0.536]$, $+0.468$ $[0.400, 0.534]$, and $+0.477$ $[0.403, 0.550]$ under public mean/std, validation-split z-score, and raw-action$/100$ preprocessing, respectively. For ridge values $10^{-3}, 10^{-2}, 10^{-1}$, the fixed-target dynamics joint R^2 is 0.479, 0.485, and 0.498, while its gain over action is $+0.467$, $+0.469$, and $+0.466$. Past-action gains remain negative at $h = 8$ (approximately -0.059 for the dynamics input), and deranged-action controls do not reproduce the positive forward readout.

The target-resolution control is essential. Predicting the complete 192-dimensional latent increment gives a dynamics-top-8 state-plus-action gain of -0.038 $[-0.059, -0.020]$ at $h = 8$. Accordingly, the evidence supports a source-defined, band-level forward-predictive diagnostic rather than a claim that the extracted subspace predicts the entire latent transition. The target remains an encoded latent quantity, and the protocol contains no physical intervention, causal action manipulation, online planner, or task-success endpoint; none of these stronger claims is made. As an explicit physical-target negative control, the separate normalized physical-displacement probe gives an incremental R^2 of -0.120 $[-0.174, -0.074]$ over the compact action summary for the dynamics top-8 predicted increment at $h = 8$. Thus the positive readout should be interpreted as encoded-latent predictive information, not as additional physical-motion prediction.

To expose the temporal structure rather than report only the selected $h = 8$ cell, we summarize the complete horizon profile in Table 51. At the fixed dynamics target, the dynamics top-8 gain is small and uncertain at $h = 1$ but becomes positive at $h = 4$ and $h = 8$. The joint future-versus-past contrast is also positive at $h = 8$, while the corresponding future-versus-deranged contrast is weaker; these are temporal-predictive controls, not causal tests. The matched-band analysis shows that the middle dynamics band is also predictive at the longer horizons. This is useful negative information: the result identifies a distributed source-defined dynamics geometry, not a claim that a single top covariance band contains all relevant motion information.

Table 51: Multi-horizon profile of the leakage-controlled forward readout. Entries are mean R^2 or paired R^2 contrasts with 95% held-out-episode bootstrap intervals. The fixed target is the source-defined dynamics top-8 increment; the matched-band row uses the middle dynamics band as both input and target. The future/past column compares joint state-plus-action features.

Input	h	Action	State only	State+action	Gain over action	Future–past (joint)
Dynamics top-8	1	$+0.045$ $[0.015, 0.076]$	$+0.085$ $[0.011, 0.147]$	$+0.102$ $[0.037, 0.155]$	$+0.056$ $[-0.013, 0.120]$	-0.015 $[-0.045, 0.015]$
Dynamics top-8	4	$+0.026$ $[-0.037, 0.090]$	$+0.493$ $[0.447, 0.535]$	$+0.489$ $[0.435, 0.542]$	$+0.462$ $[0.398, 0.536]$	$+0.014$ $[-0.011, 0.036]$
Dynamics top-8	8	$+0.016$ $[-0.044, 0.085]$	$+0.489$ $[0.438, 0.539]$	$+0.485$ $[0.425, 0.543]$	$+0.469$ $[0.402, 0.532]$	$+0.029$ $[0.002, 0.062]$
Covariance top-8	8	$+0.016$ $[-0.044, 0.085]$	-0.072 $[-0.158, 0.003]$	-0.077 $[-0.180, 0.009]$	-0.093 $[-0.180, -0.013]$	$+0.040$ $[-0.009, 0.087]$
Residual top-8	8	$+0.016$ $[-0.044, 0.085]$	-0.036 $[-0.096, 0.026]$	-0.040 $[-0.114, 0.036]$	-0.056 $[-0.120, 0.002]$	$+0.051$ $[-0.003, 0.107]$
Matched dynamics middle	8	-0.008 $[-0.046, 0.027]$	$+0.441$ $[0.395, 0.490]$	$+0.423$ $[0.376, 0.473]$	$+0.431$ $[0.380, 0.480]$	–

15.7 RANK- AND TARGET-SPECIFICITY OF LATENT DYNAMICS

The preceding forward readout establishes predictive information for a fixed source-defined dynamics target. We next test whether that information is specific to the corresponding latent geometry rather than a generic benefit of adding coordinates. Let B be a source-fitted orthonormal basis and let B_k denote its first k columns. For each external window we form

$$\Delta_h^{(B,k)} = (z_{t+h} - z_t)^\top B_k, \qquad \Delta_h^{\mathrm{phys}} = \mathrm{wrap}(s_{t+h} - s_t), \tag{S17}$$

where z is the encoded latent and s is the five-dimensional physical state. We use $k \in \{1, 2, 4, 8, 16, 32\}$ and $h \in \{1, 4, 8\}$. The four target families are the dynamics-basis top-8 increment, residual-basis top-8 increment, full 192-dimensional increment, and physical displacement. The predictors are the current coordinates in each transferred basis, observed future-action summaries, or their concatenation. Bases are fitted on the separate internal frozen export and never refit on the external windows.

Every fit leaves one complete episode out. The reported incremental R^2 is the paired, window-level difference between the joint and action-only readouts; its interval is an episode-cluster bootstrap over the same held-out windows. This paired construction preserves the action baseline and avoids subtracting two unrelated confidence intervals.

Table 52: Target-specificity controls for the transferred top-8 subspaces. Entries are paired incremental R^2 over the observed-action baseline with 95% episode-cluster bootstrap intervals. The target is fixed within each row; positive values mean that the indicated current latent coordinates add predictive information beyond the action summary.

Target	Input basis	$h = 4$	$h = 8$
Dynamics-band latent increment	Dynamics top-8	$+0.349\ [0.233, 0.461]$	$+0.255\ [0.058, 0.461]$
Dynamics-band latent increment	Covariance top-8	$-0.453\ [-0.909, -0.188]$	$-0.761\ [-1.585, -0.194]$
Dynamics-band latent increment	Random top-8	$-0.278\ [-0.509, -0.068]$	$-0.361\ [-0.579, -0.159]$
Residual-band latent increment	Residual top-8	–	$+0.279\ [0.124, 0.447]$
Residual-band latent increment	Dynamics top-8	–	$-0.255\ [-0.445, -0.100]$
Physical displacement	Dynamics top-8	$-0.098\ [-0.204, -0.027]$	$-0.209\ [-0.492, -0.022]$
Full latent increment	Dynamics top-8	$-0.238\ [-0.398, -0.101]$	$-0.294\ [-0.493, -0.129]$

The complete rank curve for the dynamics-band target is shown below. At $h = 4$, the dynamics basis becomes positive from $k = 2$ through $k = 8$, while the covariance, residual, and random controls remain negative. At $h = 8$, the same basis is positive at $k = 2, 4, 8$ but degrades when the rank is expanded, which is why the top-8 result is treated as a compact source-defined diagnostic rather than an optimal universal rank.

Table 53: Complete rank curve for the dynamics-band latent target. Cells are mean paired incremental R^2 over the observed-action baseline; the ranks were fixed before evaluating the target-specific contrasts.

	$h = 4$						$h = 8$					
Input basis	$k = 1$	2	4	8	16	32	$k = 1$	2	4	8	16	32
Dynamics	-0.050	-0.150	-0.212	$+0.349$	-1.058	-3.399	-0.002	$+0.018$	$+0.020$	$+0.255$	-0.237	-1.679
Covariance	-0.024	-0.117	-0.186	-0.453	-1.729	-8.109	-0.058	-0.075	-0.129	-0.761	-1.152	-4.288
Residual	-0.045	-0.086	-0.160	-0.370	-1.036	-3.836	-0.013	-0.034	-0.058	-0.350	-1.263	-4.152
Random	-0.052	-0.067	-0.106	-0.278	-1.152	-6.403	-0.029	-0.045	-0.151	-0.361	-0.711	-3.174

The pattern is therefore target-specific rather than a generic coordinate augmentation effect: dynamics coordinates help the dynamics-band target, residual coordinates help the residual-band target, and neither result transfers into additional physical-displacement prediction beyond observed actions. The full-latent negative control further limits the claim to source-defined latent bands. These are frozen predictive diagnostics; they do not establish a physical causal mechanism, controllability, online planning, or task success.

15.8 CROSS-HORIZON TRANSFER OF THE READOUT

The preceding rank curves fit a separate readout at each horizon. We add a strict temporal-transfer control to test whether the source-defined subspaces retain a predictive role when the readout is

estimated at one horizon and evaluated at another. For a source horizon h_s and target horizon h_t, let $\widehat{f}_{-e}^{(h_s)}$ denote a ridge readout fitted on all episodes except e using the action summary and current latent coordinates, with the target formed at h_s. We evaluate the same fitted coefficients on the held-out episode at h_t and report

$$\Gamma_B^{h_s \to h_t} = R^2\left(\widehat{f}_{B,-e}^{(h_s)}; Y_e^{(h_t)}\right) - R^2\left(\widehat{f}_{A,-e}^{(h_s)}; Y_e^{(h_t)}\right), \tag{S18}$$

where B is the current-latent basis and A is the observed-action summary. The episode is held out from both fits, and the source-fitted dynamics, residual, covariance, and random bases are never refit. We predeclare the forward transfer cells $1 \to 4$, $1 \to 8$, and $4 \to 8$; the reverse cells are retained in the machine-readable artifact because cross-horizon R^2 is not symmetric when the target scale changes.

Table 54: Cross-horizon readout transfer at rank eight. Each cell is incremental R^2 over the observed-action summary, with a 95% episode-cluster bootstrap interval. A readout is fitted at the source horizon and evaluated at the target horizon on a held-out episode.

Train → test	Target	Dynamics	Covariance	Residual	Random
$1 \to 4$	Dynamics band	$+0.083\ [0.027, 0.136]$	$-0.207\ [-0.456, -0.056]$	$-0.056\ [-0.125, 0.001]$	$-0.097\ [-0.217, 0.008]$
$1 \to 8$	Dynamics band	$+0.037\ [-0.003, 0.072]$	$-0.132\ [-0.291, -0.010]$	$-0.086\ [-0.165, -0.027]$	$-0.046\ [-0.105, 0.009]$
$4 \to 8$	Dynamics band	$+0.364\ [0.156, 0.602]$	$-0.430\ [-0.765, -0.182]$	$-0.601\ [-1.455, -0.103]$	$-0.188\ [-0.348, -0.039]$
$1 \to 4$	Residual band	$-0.030\ [-0.089, 0.035]$	$-0.065\ [-0.132, -0.014]$	$+0.041\ [0.011, 0.072]$	$-0.021\ [-0.065, 0.021]$
$1 \to 8$	Residual band	$-0.011\ [-0.036, 0.014]$	$-0.009\ [-0.028, 0.009]$	$+0.053\ [0.029, 0.076]$	$-0.008\ [-0.042, 0.026]$
$4 \to 8$	Residual band	$-0.112\ [-0.206, -0.025]$	$-0.020\ [-0.090, 0.046]$	$+0.177\ [0.108, 0.252]$	$-0.129\ [-0.261, -0.022]$

The main forward-transfer result is the $4 \to 8$ cell. For the dynamics-band target, the dynamics top-8 input exceeds the random and covariance controls by paired contrasts of $+0.553$ $[+0.351, +0.773]$ and $+0.794$ $[+0.441, +1.216]$, respectively. For the residual-band target, the residual top-8 input exceeds the dynamics input by $+0.290$ $[+0.150, +0.421]$. The same double dissociation is visible in the $1 \to 4$ and $1 \to 8$ cells, although the effects are smaller. The paired contrasts, including all nine source–target horizon combinations and the reverse direction, are included in the accompanying JSON and Markdown artifact. This cross-horizon result strengthens the interpretation as a source-defined latent predictive subspace, but it remains a frozen association beyond an observed action summary. It does not establish a physical state variable, causal action effect, online planning, task success, or retraining gain.

15.8.1 Normalization and window-density robustness

The cross-horizon result is not tied to the public action normalization or to the two-window sampling density. We repeat the complete frozen readout for public normalization, validation-split z-scoring, and raw actions divided by 100, using two, four, and up to eight windows per episode. All nine cells retain the same 21 held-out episodes, leave-one-episode-out fitting, rank-8 source bases, and 2,000 episode-cluster bootstrap replicates. Short episodes contribute fewer than eight windows when the requested density is infeasible; the resulting totals are therefore 42, 84, and 164 windows.

Table 55 reports the predeclared forward contrasts. The dynamics-minus-random contrast is positive in every cell for $1 \to 4$, $1 \to 8$, and $4 \to 8$, and every corresponding 95% episode-bootstrap lower bound is above zero. The $4 \to 8$ residual-minus-dynamics contrast is also positive in every cell. Thus the source-defined dynamics and residual coordinates retain target-specific predictive roles across the tested normalizations and sampling densities. This is evidence for robust cross-horizon transfer of a frozen latent readout, not a claim of normalization-invariant dynamics discovery or causal action influence.

Table 55: Normalization and window-density robustness of the rank-8 cross-horizon readout. Entries are paired mean contrasts with 95% episode-cluster bootstrap intervals. "D–R" is dynamics minus random for the dynamics-band target; "R–D" is residual minus dynamics for the residual-band target.

Normalization	Windows/episode	Episodes × windows	D–R $1 \to 4$	D–R $1 \to 8$	D–R $4 \to 8$	R–D $4 \to 8$
Public	2	21×42	$+0.180\ [+0.097, +0.273]$	$+0.082\ [+0.031, +0.140]$	$+0.553\ [+0.351, +0.773]$	$+0.290\ [+0.150, +0.421]$
Public	4	21×84	$+0.154\ [+0.120, +0.193]$	$+0.090\ [+0.064, +0.117]$	$+0.433\ [+0.268, +0.593]$	$+0.317\ [+0.209, +0.431]$
Public	8	21×164	$+0.160\ [+0.131, +0.188]$	$+0.120\ [+0.096, +0.144]$	$+0.458\ [+0.357, +0.556]$	$+0.219\ [+0.150, +0.290]$
Val.-split z	2	21×42	$+0.178\ [+0.097, +0.269]$	$+0.081\ [+0.030, +0.138]$	$+0.556\ [+0.349, +0.780]$	$+0.288\ [+0.151, +0.419]$
Val.-split z	4	21×84	$+0.153\ [+0.120, +0.192]$	$+0.089\ [+0.063, +0.116]$	$+0.434\ [+0.268, +0.595]$	$+0.317\ [+0.208, +0.433]$
Val.-split z	8	21×164	$+0.160\ [+0.131, +0.187]$	$+0.119\ [+0.095, +0.143]$	$+0.459\ [+0.358, +0.557]$	$+0.218\ [+0.148, +0.288]$
Raw/100	2	21×42	$+0.185\ [+0.101, +0.280]$	$+0.087\ [+0.036, +0.147]$	$+0.484\ [+0.319, +0.662]$	$+0.314\ [+0.173, +0.445]$
Raw/100	4	21×84	$+0.156\ [+0.122, +0.194]$	$+0.092\ [+0.066, +0.119]$	$+0.386\ [+0.251, +0.503]$	$+0.320\ [+0.226, +0.417]$
Raw/100	8	21×164	$+0.162\ [+0.133, +0.190]$	$+0.121\ [+0.097, +0.146]$	$+0.435\ [+0.348, +0.520]$	$+0.255\ [+0.184, +0.325]$

15.8.2 SOURCE-FIT ROBUSTNESS

The preceding matrix keeps the source-defined basis fixed to the canonical internal fit. To test whether the transfer depends on that particular fit, we repeat the most informative density settings using two independently fitted dynamics/residual basis pairs (source splits 20261006 and 20261007). The external validation episodes, model checkpoint, readout, normalization arms, and bootstrap protocol remain unchanged. Table 56 reports the complete 12-cell audit. Every forward dynamics-minus-random interval and every $4 \to 8$ residual-minus-dynamics interval has a positive lower endpoint across both source splits. The result therefore controls source-fit dependence without changing the frozen predictive interpretation.

Table 56: Source-fit robustness of the rank-8 cross-horizon readout. The dynamics and residual bases are independently fitted on two source splits; entries are paired mean contrasts with 95% episode-cluster bootstrap intervals. D–R denotes dynamics minus random for the dynamics-band target, D–C dynamics minus covariance, and R–D residual minus dynamics for the residual-band target.

Source	Normalization	Windows	Episodes × windows	D–R $1 \to 4$	D–R $1 \to 8$	D–R $4 \to 8$	D–C $4 \to 8$	R–D $4 \to 8$
20261006	Public	2	21×42	$+0.208$ $[+0.123, +0.307]$	$+0.089$ $[+0.041, +0.145]$	$+0.564$ $[+0.339, +0.834]$	$+0.854$ $[+0.417, +1.386]$	$+0.295$ $[+0.159, +0.426]$
20261006	Public	8	21×164	$+0.170$ $[+0.142, +0.199]$	$+0.124$ $[+0.099, +0.150]$	$+0.458$ $[+0.359, +0.551]$	$+0.468$ $[+0.346, +0.630]$	$+0.210$ $[+0.143, +0.276]$
20261006	Val.-split z	2	21×42	$+0.205$ $[+0.122, +0.302]$	$+0.087$ $[+0.040, +0.141]$	$+0.568$ $[+0.337, +0.845]$	$+0.863$ $[+0.431, +1.378]$	$+0.293$ $[+0.157, +0.423]$
20261006	Val.-split z	8	21×164	$+0.170$ $[+0.142, +0.198]$	$+0.123$ $[+0.099, +0.149]$	$+0.459$ $[+0.360, +0.552]$	$+0.472$ $[+0.349, +0.635]$	$+0.208$ $[+0.141, +0.275]$
20261006	Raw/100	2	21×42	$+0.214$ $[+0.127, +0.316]$	$+0.094$ $[+0.045, +0.150]$	$+0.481$ $[+0.296, +0.681]$	$+0.872$ $[+0.338, +1.623]$	$+0.325$ $[+0.183, +0.459]$
20261006	Raw/100	8	21×164	$+0.172$ $[+0.144, +0.200]$	$+0.125$ $[+0.100, +0.151]$	$+0.433$ $[+0.350, +0.510]$	$+0.447$ $[+0.330, +0.606]$	$+0.246$ $[+0.180, +0.314]$
20261007	Public	2	21×42	$+0.197$ $[+0.111, +0.295]$	$+0.089$ $[+0.033, +0.158]$	$+0.589$ $[+0.334, +0.877]$	$+0.826$ $[+0.426, +1.290]$	$+0.300$ $[+0.159, +0.445]$
20261007	Public	8	21×164	$+0.172$ $[+0.141, +0.204]$	$+0.121$ $[+0.095, +0.147]$	$+0.455$ $[+0.363, +0.542]$	$+0.479$ $[+0.353, +0.636]$	$+0.213$ $[+0.143, +0.285]$
20261007	Val.-split z	2	21×42	$+0.195$ $[+0.110, +0.290]$	$+0.087$ $[+0.032, +0.154]$	$+0.593$ $[+0.337, +0.888]$	$+0.847$ $[+0.444, +1.322]$	$+0.297$ $[+0.158, +0.441]$
20261007	Val.-split z	8	21×164	$+0.172$ $[+0.140, +0.203]$	$+0.121$ $[+0.095, +0.147]$	$+0.456$ $[+0.364, +0.544]$	$+0.484$ $[+0.356, +0.643]$	$+0.211$ $[+0.140, +0.283]$
20261007	Raw/100	2	21×42	$+0.203$ $[+0.115, +0.302]$	$+0.094$ $[+0.037, +0.164]$	$+0.502$ $[+0.291, +0.716]$	$+0.764$ $[+0.346, +1.319]$	$+0.329$ $[+0.186, +0.472]$
20261007	Raw/100	8	21×164	$+0.174$ $[+0.142, +0.205]$	$+0.123$ $[+0.096, +0.149]$	$+0.429$ $[+0.352, +0.503]$	$+0.450$ $[+0.337, +0.592]$	$+0.249$ $[+0.178, +0.320]$

15.8.3 RICHER OBSERVED-ACTION HISTORY CONTROL

The compact action summary used above is intentionally low-dimensional, but it could still leave temporal structure in the observed actions unmodeled. We therefore repeat the complete cross-horizon audit with a richer control. For each window and horizon, we retain the mean of each two-dimensional action block in temporal order. Because cross-horizon transfer requires a common feature dimension, the sequence is left-aligned and zero-padded to the prespecified maximum horizon $H = 8$, giving a 16-dimensional action-history control. The compact six-dimensional summary and this richer control use the same source-fitted bases, ridge scale, leave-one-episode-out folds, held-out episodes, and 2,000-replicate episode-cluster bootstrap.

The canonical matrix contains three action normalizations and two sampling densities; the source-fit replication contains two independent source splits with the same normalization and density arms. Thus the combined audit has 18 cells. Table 57 reports, for each cell, the minimum 95% lower endpoint across the three predeclared forward transfers ($1 \to 4$, $1 \to 8$, and $4 \to 8$). The richer-history dynamics-minus-random contrast remains positive in all cells, with a matrix minimum of $+0.0278$; the richer-history dynamics-minus-covariance minimum is $+0.0499$. For the complementary residual target, the $4 \to 8$ residual-minus-dynamics lower endpoint is positive in all 18 cells, ranging from $+0.168$ to $+0.191$.

Table 57: Robustness to a richer observed-action history. Each entry is the minimum 95% episode-cluster bootstrap lower endpoint across the three predeclared forward cells for the indicated contrast. "Compact" uses the legacy six-dimensional action summary; "History" uses all per-block action means, zero-padded to 16 dimensions. All rows use 21 held-out episodes and 2,000 bootstrap replicates.

Source	Normalization	Windows	Compact D–R	History D–R	History D–C
Canonical	Public	2	$+0.0289$	$+0.0300$	$+0.0584$
Canonical	Public	8	$+0.0961$	$+0.0998$	$+0.1002$
Canonical	Val.-split z	2	$+0.0283$	$+0.0296$	$+0.0584$
Canonical	Val.-split z	8	$+0.0958$	$+0.0998$	$+0.1002$
Canonical	Raw$/100$	2	$+0.0333$	$+0.0409$	$+0.0596$
Canonical	Raw$/100$	8	$+0.0974$	$+0.1015$	$+0.1008$
20261006	Public	2	$+0.0384$	$+0.0363$	$+0.0542$
20261006	Public	8	$+0.0993$	$+0.1026$	$+0.1007$
20261006	Val.-split z	2	$+0.0373$	$+0.0358$	$+0.0542$
20261006	Val.-split z	8	$+0.0990$	$+0.1025$	$+0.1007$
20261006	Raw$/100$	2	$+0.0431$	$+0.0476$	$+0.0531$
20261006	Raw$/100$	8	$+0.1005$	$+0.1045$	$+0.1014$
20261007	Public	2	$+0.0319$	$+0.0282$	$+0.0499$
20261007	Public	8	$+0.0955$	$+0.0994$	$+0.1068$
20261007	Val.-split z	2	$+0.0312$	$+0.0278$	$+0.0499$
20261007	Val.-split z	8	$+0.0953$	$+0.0993$	$+0.1068$
20261007	Raw$/100$	2	$+0.0358$	$+0.0409$	$+0.0506$
20261007	Raw$/100$	8	$+0.0966$	$+0.1010$	$+0.1066$

The richer control has a higher feature dimension and correspondingly larger regularized condition numbers, especially when the shorter source horizons contain more zero-padded coordinates. We therefore report the compact and richer controls side by side rather than replacing the original baseline. The result is a robustness statement about frozen latent predictive specificity beyond an observed action history. It does not identify a causal action effect, physical counterfactual, online planning gain, task success, or retraining improvement.

15.9 Closed-loop physical interventions and planning

The frozen latent audits are complemented here by a direct simulator-level intervention and a live closed-loop pilot. These experiments use the released PushT checkpoint with an explicit compatibility remapping for the legacy HuggingFace ViT parameter names; the remapped checkpoint loads all $18{,}034{,}478$ parameters with no missing or unexpected keys. The action normalization follows the public PushT convention: the simulator's relative action is the raw pixel displacement divided by 100, followed by the published mean $(-0.0087, 0.0068)$ and standard deviation $(0.2019, 0.2002)$.

Matched physical intervention. For each of 128 seeds, we first sampled one simulator episode and then reset three environments with the same variation seed, explicit initial state, and explicit goal state. The environments received an equal-norm action $+a$, $-a$, or a zero action for eight steps. The primary estimand is the paired difference in the final block pose; intervals are 2,000 pair-bootstrap replicates. This is a causal intervention inside the PushT simulator, not a claim about the learned representation.

Table 58: Matched PushT physical intervention. The plus/minus contrast is computed from identical simulator states, goals, and variations.

Estimand	Estimate	95% low	95% high
Block XY effect $\|y_+ - y_-\|$	13.202	8.131	19.147
Block pose effect $\|q_+ - q_-\|$	13.207	7.962	19.178
Goal-distance effect $(+) - (-)$	2.377	-1.527	6.735
Reward effect $(+) - (-)$	-38.601	-152.540	77.913
XY displacement, plus	7.651	descriptive	
XY displacement, minus	7.578	descriptive	
XY displacement, zero	1.308	descriptive	

The intervention produces a nonzero physical block response, while the goal-distance and reward contrasts include zero. We therefore use this result only to establish action sensitivity of the physical simulator and do not interpret it as a task-improvement result.

Held-out model counterfactual. We next tested whether the frozen predictor's action-conditional endpoint is physically predictive. A Ridge state readout was fitted on 64 calibration rollouts and evaluated on 16 disjoint intervention pairs (plus, minus, and zero), each with an eight-step horizon. The predictor shows a latent counterfactual separation of 10.117 (95% CI $[8.315, 12.093]$), while the actual physical XY separation is 14.076 (95% CI $[2.250, 31.956]$). However, the decoded physical position error is 57.962 for the plus arm, 68.716 for the minus arm, and 54.751 for zero. These errors exceed the constant-initial-pose baseline by 55.023, 57.192, and 54.121 pixels, respectively, with all corresponding bootstrap intervals strictly positive. Thus the checkpoint is action-sensitive in latent space in this pilot, but the held-out physical-state counterfactual is not validated; in particular, the latent separation must not be presented as accurate physical prediction.

Horizon sensitivity of physical counterfactuals. To test whether this gap is a one-step artifact, we repeated the matched intervention on the same 16 held-out states and action directions at horizons $h \in \{1, 2, 4, 8\}$. Each horizon received its own Ridge readout fitted on the same 64 disjoint calibration rollouts truncated to that horizon. Table 59 shows that latent endpoint error increases from 3.812 to 8.331, while the decoded physical error remains far above the constant-pose baseline at every horizon. The physical separation between $+a$ and $-a$ grows from 0.580 to 14.076 pixels, but the latent separation is not a calibrated physical prediction: it varies between 9.822 and 13.012 and does not track the physical separation monotonically.

Table 59: Matched physical counterfactual horizon sweep. The same held-out states and action directions are used at all horizons; each row has a separate calibration readout. Position errors are in pixels and intervals are episode-bootstrap intervals in the machine-readable artifact.

h	Latent endpoint error	Decoded $+a$ error	Decoded $-a$ error	Constant $+a$ error	Physical $+/-$ separation
1	3.812	41.567	44.494	0.221	0.580
2	5.925	49.157	49.675	1.194	2.658
4	7.713	43.385	58.439	2.575	7.257
8	8.331	57.962	68.716	2.938	14.076

Live online planning and normalization sensitivity. Finally, we executed the policies in the live simulator rather than replaying dataset actions. We used four fixed seeds and the same planning protocol for three action-normalization arms: identity, the public live-action z-score, and raw-action-divided-by-100 followed by the public z-score. Each arm used horizon five, receding horizon five, action block five, eight CEM candidates, and one CEM iteration. Zero, random, vanilla LeWM CEM, and spectral-cost LeWM CEM were evaluated in every arm. All twelve policy–normalization cells had zero task success. The matched means are reported in Table 60; the corresponding episode-level paired bootstrap summaries are included in the artifact record.

Table 60: Matched live PushT planning sensitivity to action normalization. Every cell uses the same four seeds and the same CEM protocol within the matrix. Lower distance is favorable; higher return is favorable; clipping is the fraction of executed action coordinates at the bound. No cell achieved task success.

Normalization	Policy	Success	Final distance	Block XY distance	Clip fraction
Identity	Zero	0.00	233.684	182.502	0.000
Identity	Vanilla CEM	0.00	437.527	167.819	0.021
Identity	Spectral CEM	0.00	397.283	197.348	0.024
Live z-score	Zero	0.00	233.684	182.502	0.000
Live z-score	Vanilla CEM	0.00	252.568	182.502	0.000
Live z-score	Spectral CEM	0.00	253.166	182.502	0.000
Raw/100 + z-score	Zero	0.00	233.684	182.502	0.000
Raw/100 + z-score	Vanilla CEM	0.00	973.074	152.880	0.911
Raw/100 + z-score	Spectral CEM	0.00	1045.815	188.628	0.908

This matched sensitivity test rules out a simple normalization-only rescue of the online claim. Identity vanilla CEM reduces block XY distance by 14.682 pixels relative to zero action but increases final state distance by 203.844 pixels; the public live z-score leaves block distance unchanged and still has zero success; and raw/100 produces severe saturation (0.911 clipping for vanilla CEM) despite a lower block-distance mean. The random control is also included in the artifact and has zero success in all three arms. We therefore report these results as negative closed-loop evidence: they establish that the current checkpoint and planner interface are executable, but do not establish online planning gain or task success. The larger evaluation remains gated on a valid PushT HDF5 dataset and a stable CUDA runtime; the currently present compressed file fails the Zstandard integrity check and is not used as evidence.

These three tests separate the estimands that are often conflated: simulator action effect, frozen-model latent counterfactual sensitivity, and closed-loop task success. The present evidence supports only the first and the latent sensitivity portion. It does not support a physical causal model, accurate physical counterfactual prediction, online planning gain, or task success.

Learned physical-dynamics closure and residual-target ablation. The preceding frozen-predictor test does not ask whether a separately learned physical adapter can recover simulator dynamics. We therefore added a strictly separated closure benchmark in `results/pusht/lewm/learned_physical_dynamics_residual_20261004`. The training data contain 256 sequential one-step transitions with a 50–50 mixture of random and block-directed behavior, so that contact events are represented. A separate 128-state rendered subset fits the frozen-LeWM embedding to an eight-dimensional physical-state feature; 16 held-out initial states and independent action directions are used only for evaluation. The learned state transition is an ExtraTrees model, and the visual arm uses the frozen embedding only to initialize that transition model. No LeWM weights are updated.

As a model-form ablation, we compared absolute next-state regression with residual regression. With the same contact-covered data, absolute regression produced $h = 1$ block-position errors of 83.864 and 83.973 for the two action arms, while residual regression reduced them to 2.362 and 3.538. This reduction is a parameterization effect, not evidence of an improved world model: the constant-pose baseline is still 0.126 and 1.352 for the same two arms because most short rollouts do not contact the block. The residual state model's predicted $+a/-a$ separation is 4.215 at $h = 1$ versus the simulator separation 1.229, and its errors grow to 30.121 and 30.178 at $h = 8$. The visual initialization is substantially worse: its $h = 1$ errors are 106.480 and 106.492, and its $h = 8$ errors are 107.567 and 116.244. The complete paired table is shown in Table 61.

Table 61: Held-out physical counterfactual closure with contact-covered training. Each entry is a mean block-XY endpoint error over 16 held-out cases. The state arm uses the true initial physical state; the visual arm initializes the same residual transition model from a frozen LeWM embedding. The constant arm repeats the initial state.

h	State $+a$	Visual $+a$	Constant $+a$	State $-a$	Visual $-a$	Constant $-a$
1	2.362	106.480	0.126	3.538	106.492	1.352
2	5.341	105.156	0.712	9.462	109.272	1.631
4	19.511	106.991	7.323	15.345	107.607	4.756
8	30.121	107.567	5.951	30.178	116.244	4.255

We then used the learned models in a matched receding-horizon MPC pilot with zero, random, a privileged heuristic, simulator-oracle candidate ranking, state-model MPC, and visual-model MPC. All six policies had zero success on the four fixed episodes. The oracle candidate ranker reduced mean final block XY distance to 166.431 versus 197.862 for zero control, but its full-state distance and return worsened because the candidate objective does not enforce a stable agent trajectory. State-model and visual-model MPC obtained block distances of 222.333 and 212.286, respectively, and final state distances of 814.632 and 1487.475. These are diagnostic negative results: simulator action response and a short-horizon learned state effect can be measured, but the frozen latent-to-physical interface and closed-loop task success remain unvalidated. This benchmark does not upgrade the frozen latent audit to learned physical causality, accurate physical counterfactuals, or planning improvement.

Renderer-aware image controls and observability decomposition. The preceding readout uses only frozen latent vectors, so a failure could in principle arise either from the image encoder or from the action-conditioned predictor. We therefore added a matched capacity control using the rendered images themselves. For each horizon, a Ridge readout was fit on the same 64 calibration endpoints used above and evaluated on the same 16 held-out paired states. The image controls were (i) a 32×32 raw-pixel vector and (ii) a renderer-aware block mask containing the block centroid, second moments, area, and the agent centroid. The latter targets the grey-blue movable block, not the light-green goal; all feature extraction and readout fitting were performed without using held-out endpoints. We compare these controls with the actual final latent, the frozen predictor's final latent, and the initial latent repeated across the three interventions.

Table 62: Image-capacity controls for frozen physical observability. Entries are mean block-position endpoint errors in pixels for the $+a$ arm over 16 held-out cases; the machine-readable artifact reports episode-bootstrap 95% intervals and the corresponding $-a$ and zero arms. The calibration set is disjoint from the held-out cases.

h	Observed latent	Predicted latent	Initial latent	Block mask	Raw pixels	Physical $+/-$ separation
1	45.884	48.394	47.105	40.396	66.771	2.167
2	43.728	57.834	44.799	45.664	66.457	7.195
4	50.985	63.822	53.532	47.599	72.376	18.045
8	57.832	58.780	59.209	45.205	78.122	45.484

The renderer-aware block readout is substantially more informative about position than the raw-pixel control, as expected for a direct image-level capacity control. The observed latent endpoint error ranges from 43.728 to 57.832 pixels across horizons, while the predicted-latent error ranges from 48.394 to 63.822 and the initial-latent control from 44.799 to 59.209. At $h = 8$, for example, the block-mask error is 45.205 $[38.088, 52.310]$, compared with 57.832 $[43.437, 73.179]$ for the observed latent and 58.780 $[45.997, 72.615]$ for the predicted latent. Thus the rendered image contains a recoverable physical position cue, whereas the 64-rollout frozen-latent readout used in this decomposition does not match that cue closely enough to serve as a validated physical-state interface. The predicted latent separation remains nonzero (9.914, 12.890, 11.809, and 11.329 for $h = 1, 2, 4, 8$), whereas the simulator's physical $+a/-a$ separation grows from 2.167 to 45.484; the two quantities are therefore not interchangeable physical counterfactuals. The held-out Pearson

point estimates between predicted latent separation and physical separation are 0.562, 0.462, 0.374, and 0.655 across the four horizons, but their episode-bootstrap intervals are wide and include zero. We treat these values as a decomposition diagnostic, not as evidence of learned physical causality, accurate physical counterfactual prediction, online planning gain, or task success.

15.10 Diagnosing and repairing the negative physical closure

The initial negative closure contains two separable sources of error: the physical readout and the action-conditioned transition model. We tested the first source by holding the 24-case held-out intervention set fixed and sweeping the number of disjoint calibration endpoints. The target was reduced to block XY for the position readout; full-pose and nonlinear decoders were additional controls.

Table 63: Calibration-size and decoder-class ablation at $h = 8$. Entries are mean block-XY errors in pixels on the same 72 held-out intervention images (24 cases times $+a$, $-a$, and zero). The seed-6172 artifact is the machine-readable source; all intervals are episode-bootstrap intervals.

Calibration size	Latent Ridge observed	Latent Ridge predicted	Latent Ridge initial	ExtraTrees observed	Block mask observed	Raw pixels observed
128	39.799	58.365	43.576	59.737	41.175	52.976
256	22.278	43.577	25.036	47.984	41.301	55.501
512	19.062	42.858	23.656	43.719	40.189	55.184

Calibration coverage materially changes the observed-latent readout: the latent Ridge error falls from 39.799 at 128 endpoints to 19.062 at 512, whereas ExtraTrees remains worse at 43.719. The predicted-latent error is still 42.858 at 512, so improving the readout does not make the frozen predictor's counterfactual endpoint physically accurate. The result changes the interpretation of the earlier failure: the frozen latent contains recoverable position information under a sufficiently covered linear readout, but its predicted endpoint does not preserve that information reliably.

The second source is the state interface supplied to the learned physical transition. The simulator state is agent XY, block XY, block angle, and agent velocity. A single rendered frame does not identify instantaneous velocity, so we compared a pose-only closure (agent XY, block XY, and block angle) with the earlier full-state interface. With 512 transition samples and a matched visual calibration set, pose-only visual endpoint errors for the $+a/-a$ arms were $16.031/17.662$ at $h = 1$ and $41.257/44.042$ at $h = 8$, compared with roughly 44–76 pixels for the earlier full-state visual arm. A two-frame latent adapter reduced velocity error from 153.619 to 132.565 and full-state error from 180.463 to 164.424, but increased block-XY error from 19.147 to 22.571. Thus temporal features recover part of the velocity signal without providing a uniformly better physical interface.

The repaired pose interface improves the physical readout but does not solve the downstream task: the matched pose-only visual MPC and state MPC both have zero success on the four fixed episodes, and the zero-action policy remains at the same block-distance baseline. The remaining failure is therefore consistent with contact-mode transition error and an underconstrained planner objective, not with a complete absence of physical information in the frozen latent. These follow-up results are diagnostic of interface and model-form sensitivity; they do not establish learned physical causality, accurate physical counterfactual prediction, online planning gain, or task success.

To identify the remaining closed-loop bottleneck, we replayed 64 identical candidate actions from each of 16 held-out states through both the pose model and the simulator. The learned-model ranking for block-only distance has Spearman correlation 0.063 $[0.004, 0.125]$ with the simulator ranking and incurs 8.901 pixels of selected regret. A task cost that also includes agent position and block angle produces a much higher ranking correlation of 0.974 $[0.961, 0.984]$, but the best candidate in this one-block random set still has mean task cost 207.379. This separates two remaining failure modes: block-only ranking is a poor contact objective for the learned pose model, and the sampled action horizon does not reliably contain a successful multi-stage push. The diagnostic motivates a contact-aware, multi-stage candidate protocol; it is not itself evidence of online planning success.

Contact-aware repair and objective ablation. We next audited and repaired the simulator-state diagnostic while keeping it separate from the frozen-LeWM visual interface. The audit found that PushT's `n_contacts` aggregates wall and block collisions under the same collision type; it is therefore not a valid agent–block contact label. The revised label is block-pose change after the transition, with wrapped angular differences. We also retain the simulator's agent velocity, use the

actual post-reset observation as the model input, and match the variation seed across state resets. Finally, the repaired protocol replaces hand-placed contact rings with genuine free, approach, and tangential transitions sampled along short simulator rollouts. This directly tests whether the earlier negative result was caused by an invalid label or by a contact-distribution mismatch.

Table 64: Repaired contact-aware physical-adapter diagnostic on 48 held-out cases for each independent seed. Entries are mean block-XY errors in pixels; episode-bootstrap 95% CIs are available in the machine-readable artifacts. Contact is defined by block-pose change, not the environment's aggregate collision counter. The no-block-motion stratum is closed, while contact-mode transitions remain the dominant error source.

Seed	Horizon	Overall	Constant	Block-motion error	n_{motion}	No-motion error	$n_{\text{no motion}}$
7317	1	1.821	1.828	7.946	11	0.000	37
7317	4	6.870	7.640	21.985	15	0.000	33
7317	8	12.185	15.224	38.991	15	0.000	33
9173	1	1.331	1.186	8.327	7	0.000	41
9173	4	4.269	4.148	18.628	11	0.000	37
9173	8	6.697	8.554	24.727	13	0.000	35

The corrected gate eliminates predicted motion on the no-block-motion stratum, but the dominant error remains contact-mode transition: at $h = 8$ the block-motion errors are 38.991 and 24.727 pixels across the two seeds, despite lower unconditional error than the constant baseline in both seeds. This identifies the surviving bottleneck as contact dynamics and mode coverage, not generic XY representation. In the matched eight-episode closed-loop pilot, the best two-block block-aware MPC reached 86.354 and 177.159 pixels of block XY distance for seeds 7317 and 9173, respectively; the corresponding uniform-MPC distances were 248.602 and 178.660 pixels, and the privileged heuristic distances were 248.995 and 278.608 pixels. However, every policy had zero full-task successes and zero block-pose successes in both seeds. Expanding the candidate horizon to four blocks did not improve the result (mean block distance 164.636 in the same seed), so the longer candidate protocol is not adopted as a rescue. These are simulator-state adapter diagnostics: they neither retrain LeWM nor validate physical counterfactuals, online planning gain, or task success from the frozen latent. The machine-readable sources are `results/pusht/lewm/contact_velocity_trajectory_final_7317_20261004/` and `results/pusht/lewm/contact_velocity_trajectory_final_9173_20261004/`.

Contact-conditioned predicted-latent adapter. The repaired simulator-state protocol does not by itself explain whether the frozen predictor preserves the contact effect. We therefore held the encoder fixed, sampled genuine approach-to-contact states, and used disjoint calibration and held-out seeds. Each case received matched outward-normal $+a$, $-a$, and zero interventions; contact was verified by block-pose change. For every horizon, we compared a readout calibrated on predicted endpoint latents, an action-conditioned latent transition adapter initialized from the current latent, an observed-final-latent upper-bound readout, and the constant-pose baseline. The action-conditioned adapter is a downstream diagnostic, not a retrained LeWM predictor.

Table 65: Contact-conditioned physical-readout repair. Entries are mean $+a$ block-XY endpoint errors in pixels. Each row uses disjoint calibration and held-out contact cases; contact rates were 1.00 in both splits. The direct action-latent adapter uses the frozen current latent and the proposed action, whereas the observed-latent column is an upper-bound interface that encodes the actual final image.

Split	h	Predicted latent	Direct action–latent	Observed latent	Constant	Physical $+/-$	Predicted latent $+/-$
8421	1	50.464	35.731	32.905	31.197	26.257	15.084
8421	2	46.951	36.715	26.713	60.981	56.120	16.704
8421	4	46.285	55.248	37.384	105.081	100.447	14.720
8421	8	67.666	68.794	55.783	121.935	117.534	15.870
9271	1	36.768	22.412	20.220	32.978	27.641	13.269
9271	2	41.945	29.708	20.239	62.705	57.425	14.546
9271	4	48.733	49.206	26.381	102.107	96.965	12.755
9271	8	61.343	69.165	35.376	122.095	117.577	14.034

The direct action-latent adapter improves on the constant baseline at $h = 2, 4, 8$ in both splits, while the predicted endpoint readout alone is less stable and the observed-final-latent readout remains better. Crucially, the frozen predictor's latent $+a/-a$ separation stays near 13–17 pixels while the physical separation reaches roughly 56–118 pixels at the same horizons. Thus the contact and calibration repairs close part of the downstream interface gap, but they cannot reconstruct an action effect that the original predictor does not preserve. The evidence supports a precise repair target—action-sensitive predictor training or an explicitly learned physical transition adapter—and does not support accurate physical counterfactuals or online planning for the unmodified frozen checkpoint.

Teacher-forced multi-step predictor repair. The preceding adapter repairs the downstream physical readout but leaves the released action-conditioned predictor unchanged. We therefore tested whether autoregressive exposure bias accounts for the remaining action-effect compression. For each genuine approach–contact case, we collected the full simulator trajectory under matched $+a$, $-a$, and zero interventions. The visual encoder and projector were frozen. Only the action encoder, predictor, and prediction projection were updated, using the frozen encoder's latent at every future simulator step as a teacher-forced target. At test time, the repair was evaluated autoregressively, with no target latent supplied to the history. The public PushT action protocol was applied after dividing live actions by the environment's action scale of 100. The two independent runs used 48 training cases (144 intervention trajectories) and 16 held-out cases (48 paired intervention rows) each; calibration and test cases were disjoint.

Table 66 reports the $+a$ block-XY error. The base column uses a readout calibrated on the base predictor distribution; the repaired-calibrated column uses a separate readout calibrated on the repaired predictor distribution. The paired gain is defined as base error minus repaired-calibrated error, so positive values favor the repair. The native gain applies the original base readout to the repaired latent and is a deliberate readout-shift control.

Table 66: Teacher-forced contact-rich predictor repair. Entries are mean $+a$ block-XY errors in pixels; paired gains are positive when the repaired calibrated readout is better. Intervals are episode-bootstrap 95% intervals over the 16 held-out contact cases and are kept separate by seed. Native gain uses the original base readout and exposes latent-distribution shift.

Seed	h	Base error	Repaired calibrated error	Paired gain	Gain 95% CI	Native gain
8421	1	45.537	45.256	$+0.281$	$[-7.144, +7.428]$	-2.014
8421	2	60.758	53.796	$+6.962$	$[-5.417, +18.944]$	-24.607
8421	4	109.074	76.266	$+32.807$	$[+7.936, +60.540]$	-90.619
8421	8	112.480	88.859	$+23.620$	$[-5.709, +54.881]$	-136.365
9271	1	51.589	51.153	$+0.437$	$[-6.137, +6.087]$	-4.319
9271	2	77.144	63.388	$+13.756$	$[-1.210, +28.762]$	-6.903
9271	4	109.446	72.431	$+37.015$	$[+15.548, +63.198]$	-23.361
9271	8	125.492	95.770	$+29.722$	$[-2.044, +59.697]$	-53.798

The point estimate improves in both seeds at $h = 2, 4, 8$, with a clearly positive paired interval at $h = 4$ in both seeds; the $h = 2$ and $h = 8$ intervals remain compatible with zero. The repaired latent separation also increases substantially, but its native readout becomes worse, so the separation alone is not a physical-accuracy claim. The result supports a specific, partially successful repair of the diagnosed action-effect compression and identifies readout recalibration as necessary. It does not show that the original frozen checkpoint already had this behavior, and it does not establish learned physical causality, accurate physical counterfactuals, online-planning gain, or task success. Machine-readable outputs are stored under `results/pusht/lewm/contact_predictor_teacher_forced_20261004_seed8421/`, `results/pusht/lewm/contact_predictor_teacher_forced_20261004_seed9271/`, and the seed-separated summary under `results/pusht/lewm/contact_predictor_teacher_forced_summary_20261004/`.

Three-arm audit and observed-latent upper bound. The $+a$ table above is not sufficient to distinguish an action effect from an action-independent readout bias. We therefore retained the matched $-a$ and zero arms in the extended evaluation. We also fit a separate decoder from the observed target latent itself to the physical pose. This column is an interface upper bound for the

available rendered observation and is not a counterfactual predictor: it sees the actual future target image. Table 67 reports point estimates for all arms. The physical $+a/-a$ separation is reported alongside the arm-wise gains so that latent sensitivity is not mistaken for physical accuracy.

Table 67: Extended teacher-forced repair audit. Errors are block-XY pixels; the observed-latent column is fit to the actual target latent and therefore serves only as an observation/readout upper bound. Gains are base error minus repaired-calibrated error, so positive values favor the repair.

Seed	h	Base $+a$	Repaired $+a$	Observed-latent $+a$	Gain $+a$	Gain $-a$	Gain zero	Physical $+/-$
8421	1	45.537	45.256	29.808	$+0.281$	$+1.429$	$+0.220$	26.257
8421	2	60.759	53.795	26.107	$+6.963$	$+8.694$	$+8.734$	56.120
8421	4	109.074	76.265	39.601	$+32.809$	$+36.330$	$+40.065$	100.447
8421	8	112.479	88.859	68.100	$+23.620$	$+41.709$	$+43.829$	117.534
9271	1	51.589	51.153	30.844	$+0.436$	-1.335	-0.863	26.633
9271	2	77.144	63.387	30.566	$+13.757$	$+3.871$	$+5.087$	56.135
9271	4	109.446	72.431	49.483	$+37.014$	$+14.801$	$+16.303$	98.430
9271	8	125.492	95.772	43.274	$+29.720$	$+37.132$	$+27.732$	122.174

At $h = 4$, all three arm-wise point gains are positive for both seeds, while the observed-latent error remains far below the predictor-based error. This separates a repaired action-conditioned transition interface from the remaining visual/physical observability gap. The $h = 1$ rows also show why a single short-horizon result is not decisive: the effect is small and its intervals include zero.

Calibration-size and training-strength ablations. The repaired predictor changes the latent distribution, so calibration size must be measured independently of predictor fitting. With the predictor fixed at the eight-epoch repair, the $h = 4$ $+a$ error decreased from roughly $167.8/132.6$ pixels with eight calibration cases to $76.3/72.4$ pixels with 48 cases for seeds 8421/9271; the intermediate 16- and 32-case values are retained in Table 68. These are calibration-only changes, not additional predictor training.

Table 68: Calibration-size ablation at $h = 4$. The repair and held-out cases are fixed; only the number of disjoint training cases used to fit the repaired physical readout changes.

Calibration cases	Seed 8421 $+a$ error	Seed 9271 $+a$ error	Interpretation
8	167.775	132.574	under-calibrated
16	124.590	105.439	unstable
32	83.262	118.844	approaching regime
48	76.265	72.431	full calibration

The same protocol also gives a one-seed sensitivity audit for repair strength. At $h = 4$, one, eight, and 32 epochs yielded $+10.683$, $+32.809$, and $+53.722$ pixel paired gains, respectively; anchor weights 0, 0.05, and 0.2 yielded $+35.919$, $+32.809$, and $+27.322$. With only 16 training cases the point estimate was negative (-30.839), whereas 32 cases yielded $+29.684$. Table 69 reports the full intervals. Because these rows use one seed, they are sensitivity diagnostics, not a new multi-seed ranking claim.

Table 69: Single-seed training-strength and sample-size ablations at $h = 4$. The held-out contact cases are fixed. Intervals are episode-bootstrap 95% intervals; positive paired gains favor the repaired calibrated readout.

Setting	Training cases	Base $+a$ error	Repaired $+a$ error	Gain [95% CI]
1 epoch	48	109.074	98.392	$+10.683$ [-11.202, $+33.080$]
8 epochs	48	109.074	76.265	$+32.809$ [$+7.938$, $+60.542$]
32 epochs	48	109.074	55.353	$+53.722$ [$+22.431$, $+86.981$]
anchor $= 0$	48	109.074	73.155	$+35.919$ [$+11.503$, $+63.126$]
anchor $= 0.2$	48	109.074	81.752	$+27.322$ [$+2.183$, $+52.626$]
16 training cases	16	115.120	145.959	-30.839 [-68.610, $+4.942$]
32 training cases	32	128.365	98.681	$+29.684$ [-9.966, $+78.361$]

These ablations explain the original negative result without overclaiming a complete solution. Correct action scaling and contact context make the intervention identifiable; teacher-forced multi-step fitting recovers part of the action effect; and sufficient calibration data are required

to read the repaired latent. The observed-latent ceiling and the wide intervals at some horizons show that the remaining gap is not merely an optimization artifact. None of these targeted repairs establishes learned physical causality, accurate physical counterfactuals, online planning gain, or task success for the unmodified frozen checkpoint. Machine-readable outputs are stored under `results/pusht/lewm/contact_predictor_deep_20261005_seed8421/`, `results/pusht/lewm/contact_predictor_deep_20261005_seed9271/`, `results/pusht/lewm/contact_predictor_deep_summary_20261005/`, and `results/pusht/lewm/contact_predictor_ablation_summary_20261005/`.

Enlarged replication with 96 training and 24 held-out cases. The preceding table uses 48 training and 16 held-out contact cases. To test whether its $h = 4$ result was driven by the relatively small calibration and training sample, we repeated the same frozen-encoder, three-arm, autoregressive protocol with 96 training and 24 held-out cases for each seed. The held-out cases and the calibration endpoints were disjoint, and all live actions were again divided by the PushT action scale of 100. Table 70 reports the complete $h = 4$ and $h = 8$ audit. The observed-latent column remains an observation/readout upper bound rather than a counterfactual prediction.

Table 70: Enlarged two-seed replication of the teacher-forced predictor repair. Errors are $+a$ block-XY pixels; gains are base error minus repaired-calibrated error. The three arm columns are point gains for matched $-a$ and zero interventions. Intervals are episode-bootstrap 95% intervals over 24 held-out contact cases.

Seed	h	Base $+a$	Repaired $+a$	Observed latent $+a$	Gain $+a$	Gain 95% CI	Gain $-a$	Gain zero
8421	4	89.240	51.529	30.774	$+37.711$	$[+19.572, +55.555]$	$+32.646$	$+39.959$
8421	8	89.558	70.360	38.775	$+19.198$	$[-15.584, +48.035]$	$+2.659$	$+17.386$
9271	4	72.688	41.641	30.040	$+31.047$	$[+18.726, +44.464]$	$+29.776$	$+22.924$
9271	8	101.692	61.425	36.762	$+40.266$	$[+18.861, +65.283]$	$+19.421$	$+34.054$

At $h = 4$, the enlarged replication preserves positive point gains for all three matched arms in both seeds, and both $+a$ intervals exclude zero. This rules out the earlier negative result being explained solely by a small held-out set or by an action-independent improvement on the positive arm. At $h = 8$, the point estimates remain positive and three-arm gains remain positive, but the interval for seed 8421 includes zero. The remaining error is also visibly above the observed-latent upper bound. Thus the larger experiment strengthens the claim that contact context, correct action scaling, and teacher-forced multi-step fitting repair a substantial short-horizon interface failure; it does not close the long-horizon observability/dynamics gap or establish physical causality, accurate counterfactuals, online planning, or task success. The seed-level machine-readable outputs are stored under `results/pusht/lewm/contact_predictor_deep96_20261005_seed8421/` and `results/pusht/lewm/contact_predictor_deep96_20261005_seed9271/`.

Three-seed training-strength audit. The preceding enlarged replication used eight training epochs. To determine whether its horizon dependence was merely an optimization artifact, we repeated the same 96/24 protocol for 32 epochs at a third independent seed. This is a training-strength sensitivity audit: the visual encoder and projector, contact protocol, learning rate, anchor, calibration cases, and autoregressive evaluation are unchanged. Table 71 keeps all seed-level intervals separate and reports both horizons at which the repair is most and least stable.

Table 71: Three-seed 32-epoch training-strength audit under the 96/24 protocol. Errors are $+a$ block-XY pixels; gains are base error minus repaired-calibrated error. Intervals are episode-bootstrap 95% intervals over 24 held-out cases and are not pooled across seeds.

Seed	h	Base $+a$	Repaired $+a$	Observed latent $+a$	Gain $+a$	Gain 95% CI	Gain $-a$	Gain zero
8421	4	89.239	42.671	30.775	$+46.568$	$[+27.131, +65.197]$	$+40.598$	$+44.571$
8421	8	89.558	69.214	38.776	$+20.344$	$[-3.763, +42.373]$	$+40.931$	$+37.423$
9271	4	72.689	40.551	30.041	$+32.138$	$[+18.994, +46.797]$	$+25.513$	$+25.464$
9271	8	101.691	51.451	36.762	$+50.240$	$[+28.987, +73.636]$	$+32.647$	$+33.947$
13579	4	84.115	46.898	42.224	$+37.217$	$[+15.720, +59.176]$	$+47.649$	$+51.543$
13579	8	91.302	65.226	42.976	$+26.076$	$[-1.761, +54.744]$	$+39.921$	$+33.650$

At $h = 4$, all three seeds retain positive $+a$, $-a$, and zero point gains, and every $+a$ interval excludes zero. At $h = 8$, all point estimates remain positive, but two of the three intervals include zero. Notably, for seed 8421 the repaired latent endpoint MSE falls from 0.748 at eight epochs to 0.307 at 32 epochs, while the physical $+a$ interval still includes zero. Therefore additional optimization improves latent prediction without guaranteeing physical readout recovery; the surviving limitation is contact-mode/physical observability generalization. These results support a stable short-horizon targeted repair, not a complete long-horizon physical world model or online-planning claim. Machine-readable outputs are stored under `results/pusht/lewm/contact_predictor_deep32_20261005_seed8421/`, `results/pusht/lewm/contact_predictor_deep32_20261005_seed9271/`, `results/pusht/lewm/contact_predictor_deep32_20261005_seed13579/`, and `results/pusht/lewm/contact_predictor_deep32_summary_20261005/`.

Contact-mode coverage scaling. The three-seed audit showed that longer-horizon latent fitting alone was not sufficient: $h = 8$ remained sensitive to the physical readout and to the contact-mode distribution. We therefore held the optimizer, action protocol, contact context, and autoregressive evaluation fixed while doubling the training set from 96 to 192 contact cases and the disjoint test set from 24 to 48 cases. This is a coverage test rather than a best-of-seed selection; the enlarged run uses the same 32 epochs and 0.05 anchor as the preceding audit. Table 72 reports the two-seed replication at the horizons most diagnostic of the original failure.

Table 72: Contact-mode coverage scaling under the 192/48 protocol. Errors are $+a$ block-XY pixels; gains are base error minus repaired-calibrated error. Intervals are episode-bootstrap 95% intervals over 48 held-out contact cases. The $-a$ and zero columns are matched-arm point gains.

Seed	h	Base $+a$	Repaired $+a$	Observed latent $+a$	Gain $+a$	Gain 95% CI	Gain $-a$	Gain zero
8421	4	89.694	42.993	27.928	$+46.701$	$[+29.647, +65.638]$	$+41.843$	$+42.566$
8421	8	98.911	57.499	37.428	$+41.412$	$[+22.044, +62.088]$	$+50.614$	$+45.316$
9271	4	78.679	38.508	25.221	$+40.171$	$[+29.386, +52.135]$	$+21.955$	$+26.034$
9271	8	89.331	49.678	28.908	$+39.654$	$[+23.293, +60.053]$	$+28.163$	$+29.054$

Both independent runs close the $h = 8$ interval while preserving positive matched-arm gains, unlike the 96/24 seed-8421 run. The result supports contact-mode coverage as a causal explanation of the earlier diagnostic failure within the targeted repair protocol. It still does not establish that the released frozen checkpoint has physical causality, accurate counterfactuals, online planning gain, or task success.

Compositional action coverage and the closed-loop bridge. The constant-arm coverage experiment does not match the distribution ranked by an MPC planner: its training interventions repeat one action for the whole horizon, whereas MPC evaluates sequences that change direction and magnitude. We therefore added a separate 96/24 contact split with five action-sequence arms per case: positive and negative constant pushes, zero action, piecewise noisy pushes, and piecewise random actions. The encoder and projector remained frozen, and the repaired predictor was evaluated autoregressively. On the disjoint held-out split, the $h = 8$ latent endpoint MSE decreased from 0.860 for the original predictor to 0.381 after compositional repair.

To test whether this improvement transfers to planning, we generated the same candidate array for both predictors and replayed every candidate from the identical simulator state and goal. The selected candidate was compared with the oracle best candidate in that finite set; task cost uses the PushT termination geometry (agent position, block position, and orientation).

Table 73: Matched held-out candidate ranking after compositional repair. Each row aggregates 16 held-out contact states with 64 identical candidates per state; intervals are state-bootstrap 95% intervals. Lower selected regret is better. The oracle column is a candidate-coverage diagnostic, not a learned model result.

Predictor	Spearman	Selected regret	Oracle best	Selected actual
Original predictor	0.107 $[-0.135, 0.349]$	188.080 $[133.575, 239.737]$	169.611	357.691
Compositional repair	0.690 $[0.557, 0.797]$	32.902 $[16.053, 57.427]$	169.611	202.513

The repaired predictor therefore improves the ranking interface substantially: its selected regret is much closer to the oracle gap, while the original predictor is nearly uncorrelated with actual candidate cost. However, the oracle candidate itself remains far from the success region. Increasing the candidate count from 64 to 512 reduced the oracle task cost only from 169.6 to 148.9, while repaired ranking remained stable (0.683 Spearman and 42.6 selected regret). This identifies finite candidate coverage—including the joint agent–block–orientation requirement—as the remaining bottleneck for the tested online protocol.

Table 74: Closed-loop transfer under the compositional repair. The approach phase is a fixed state-guided contact bootstrap and is reported separately; success is the simulator's true termination flag.

Policy	Contact rate	Success	Block XY distance	Final state distance
Original MPC	1.000	0.000	152.400	1164.154
Compositional repaired MPC	1.000	0.000	107.597	659.595
Repaired MPC, privileged state/block diagnostic	1.000	0.000	79.719	737.226

The live results are deliberately bounded. They show that contact-rich, compositional repair transfers to substantially better candidate ranking and lower mean block distance, but they do not establish online task success. The privileged-state row is included only to rule out current-state visual readout as the sole explanation; it is not a vision or generalization result. The machine-readable artifacts are stored under `results/pusht/lewm/repaired_predictor_mpc_compositional_96x24_20261005/`, `results/pusht/lewm/repaired_predictor_mpc_compositional_priv_block_20261005/`, `results/pusht/lewm/repaired_predictor_candidate_rank_96x24_20261005/`, and `results/pusht/lewm/repaired_predictor_candidate_rank_512_20261005/`.

15.11 Stable contact and multistage candidate coverage

The preceding candidate-ranking result still used a first-motion contact bootstrap: the episode was handed to the planner as soon as any block pose change was detected. This event is not equivalent to a settled contact state. An agent can have high residual velocity and can already have passed the block when the first displacement is observed. We therefore added a privileged simulator-state audit with two explicit bootstrap conditions. The first-motion condition reproduces the earlier protocol; the stable-contact condition continues with a low-gain approach and a zero-target braking phase, and returns only after the agent is within 65 pixels of the block, its speed is below 80 pixels per control interval, and the most recent block displacement is negligible. This is a diagnostic bootstrap, not a visual policy.

We also replaced the mostly stationary proposal bank with a staged bank. Its templates enumerate approach, brake, contact push, short push, and a small angular fan around the desired block displacement. Every candidate is replayed from the same simulator state and is scored with the true PushT termination geometry; a simulator-in-the-loop CEM bank is reported only as an upper-bound diagnostic.

Table 75: Stable-contact candidate coverage at an eight-step horizon. The oracle column is not available to LeWM; it measures the best candidate found by replay. "Block progress" is the reduction in block-to-goal XY distance of the best candidate. No candidate reached the full PushT termination region.

Candidate set	Best task cost	Block progress	Full success	n
Uniform	273.318	48.401	0.000	16
Legacy guided	271.865	49.217	0.000	16
Staged	264.421	60.231	0.000	16
Simulator oracle	245.322	n/a	0.000	16

The staged bank therefore improves local block motion rather than solving the task. Extending the open-loop horizon to 32 steps did not close the gap: the staged bank achieved mean best block progress 141.284 and mean best block XY distance 144.754, while both full-task success and block-pose success remained zero on eight held-out states. The simulator oracle also had zero block-pose successes (mean best block XY distance 165.998). This rules out the narrow interpretation that

the eight-step endpoint alone was the only issue; contact geometry and orientation control remain unresolved.

Finally, we ran the repaired predictor with stable contact and staged proposals in the live simulator. With privileged current state, the base predictor, compositional repair, and heuristic controls all had zero success on four episodes. The repaired predictor reduced selected endpoint pose error from 492.507 to 136.858 and final state distance from 284.533 to 176.457, but its mean block distance was 108.142 versus 68.707 for the base predictor. With the visual state readout, the repaired row remained at zero success and degraded to endpoint error 422.242 and final distance 652.398. Thus the stable bootstrap removes one protocol confound and the staged bank improves a local endpoint interface, but neither result establishes physical counterfactual accuracy, online planning gain, or task success.

The machine-readable coverage audit is stored under `results/pusht/lewm/staged_candidate_coverage_20261005/` and `results/pusht/lewm/staged_candidate_coverage_h32_20261005/`; the two live runs are stored under `results/pusht/lewm/repaired_predictor_mpc_stable_staged_20261005/` and `results/pusht/lewm/repaired_predictor_mpc_stable_staged_visual_20261005/`.

15.12 Cross-experiment joint-success audit

The preceding experiments report different estimands, so a positive result on one geometry diagnostic must not be silently promoted to a functional claim. We therefore perform a final audit over the completed artifact set. The audit keeps each benchmark separate, pairs conditions by random seed within a benchmark, and does not pool observations or confidence intervals across benchmarks. We classify an intervention as "geometry-only" when every listed geometry metric improves for every paired seed but at least one functional metric does not. "All-metric joint success" is deliberately strict: every listed geometry and functional metric must improve for every paired seed.

Table 76: Strict joint-success audit across alignment and equivariance interventions. The factorial vector-world rows use predictor-equivariance and operator-alignment diagnostics together with one-step error, $h = 8$ rollout, and action composition. The checkpoint-matched nonlinear rows additionally include state-shift composition and the shared-candidate planning proxy. A "geometry-only" label is not a performance claim.

Artifact	Comparison	Geometry all seeds better	Functional all seeds better	Classification
Joint factorial	Predictor only vs none	No	No	Mixed or negative
Joint factorial	Alignment only vs none	No	No	Mixed or negative
Joint factorial	Predictor + alignment vs none	No	No	Mixed or negative
Matched nonlinear	Global alignment, $w = 0.1$	No	No	Mixed or negative
Matched nonlinear	Conditional alignment, $w = 0.01$	Yes	No	Geometry-only
Matched nonlinear	Conditional alignment, $w = 0.1$	Yes	No	Geometry-only

This audit sharpens the interpretation of the conditional phase sweep. At $w = 0.01$, conditional alignment reduces the local commutator and improves both ID and state-shift $h = 8$ rollout, but its planning-proxy regret increases and its composition result is not uniformly better. At $w = 0.1$, the same geometry–function gap becomes larger. Thus, the commutator is a useful diagnostic of a particular local geometry, but it is not a sufficient training objective for long-horizon functionality.

The same logic applies to multi-horizon training. Table 77 reports paired mean deltas relative to one-step training; negative values favor the multi-horizon profile because all entries are errors or regret. The profiles consistently improve ID long-horizon rollout, but every profile pays an ID $h = 1$ cost and incurs an OOD planning-regret cost at $h = 4$. Curriculum has the most favorable ID profile in this controlled sweep, yet it still does not dominate across domains and horizons.

Table 77: Paired multi-horizon loss audit. Entries are mean target-minus-one-step deltas over ten matched seeds. Negative values are favorable. This is a nonlinear synthetic training proxy, not an end-to-end LeWorldModel or PushT result.

Profile	ID $h = 1$	ID $h = 4$	ID $h = 8$	OOD planning $h = 4$	OOD planning $h = 8$
Uniform	$+1.768 \times 10^{-3}$	-2.645×10^{-3}	-2.245×10^{-3}	$+4.517 \times 10^{-2}$	-1.480×10^{-2}
Short-favoring	$+6.671 \times 10^{-4}$	-2.314×10^{-3}	-1.966×10^{-3}	$+2.661 \times 10^{-2}$	-1.449×10^{-2}
Long-favoring	$+3.324 \times 10^{-3}$	-2.731×10^{-3}	-2.366×10^{-3}	$+6.842 \times 10^{-2}$	-1.358×10^{-2}
Curriculum	$+1.052 \times 10^{-3}$	-2.849×10^{-3}	-2.275×10^{-3}	$+4.330 \times 10^{-2}$	-1.710×10^{-2}

To test whether this trade-off can be repaired by supervising composition directly, we add a matched composition-aware condition that gives extra weight to h=2 and h=4 prefixes containing at least one action switch. Relative to uniform multi-horizon training, the switch-minus-constant error gap decreases by -5.338×10^{-4} with a 95% bootstrap interval $[-6.616 \times 10^{-4}, -4.074 \times 10^{-4}]$. This diagnostic improvement is not a functional improvement: switch error itself increases by 1.007×10^{-4}, OOD $h = 4$ error increases by 7.318×10^{-4}, and OOD $h = 8$ error increases by 6.434×10^{-5}. The ID $h = 8$ delta is effectively zero (-4.14×10^{-7}; interval $[-8.46 \times 10^{-6}, 9.07 \times 10^{-6}]$). Thus, explicit composition weighting changes the decomposition of errors but does not convert the synthetic long-horizon trade-off into a uniform gain.

Table 78: Composition-aware training contrast against uniform multi-horizon training. Entries are paired mean deltas over ten seeds; negative values favor the composition-aware condition. The switch-gap diagnostic and functional errors are intentionally reported together.

Metric	Mean delta	95% bootstrap interval	Fraction better
Switch $h = 4$ MSE	$+1.007 \times 10^{-4}$	$[+7.38 \times 10^{-5}, +1.33 \times 10^{-4}]$	0.00
Constant $h = 4$ MSE	$+6.344 \times 10^{-4}$	$[+4.91 \times 10^{-4}, +7.91 \times 10^{-4}]$	0.00
Switch-minus-constant gap	-5.338×10^{-4}	$[-6.62 \times 10^{-4}, -4.07 \times 10^{-4}]$	1.00
ID rollout $h = 8$ MSE	-4.14×10^{-7}	$[-8.46 \times 10^{-6}, +9.07 \times 10^{-6}]$	0.70
OOD rollout $h = 4$ MSE	$+7.318 \times 10^{-4}$	$[+4.46 \times 10^{-4}, +1.03 \times 10^{-3}]$	0.20
OOD rollout $h = 8$ MSE	$+6.434 \times 10^{-5}$	$[+3.19 \times 10^{-5}, +9.34 \times 10^{-5}]$	0.20

The resulting claim matrix is intentionally asymmetric. The external frozen PushT transport result supports a robust, target-specific latent-transport claim across the three declared action normalizations and the tested window densities. Independently, the frozen dynamics/residual audit does not support additional physical-motion prediction beyond observed actions. Taken together, the evidence supports extractable and cross-horizon transportable latent dynamics information within the frozen readout protocol, but not a causal dynamics mechanism, universal planning gain, or all-metric benefit from alignment.